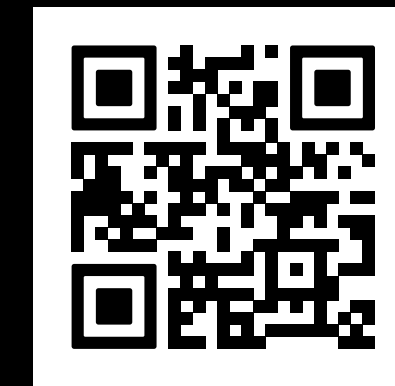


PhantomWiki: On-Demand Datasets for Reasoning & Retrieval Evaluation



arxiv.org/abs/
2502.20377



github.com/kilian-group/
phantom-wiki



Albert Gong*



Chao Wan*



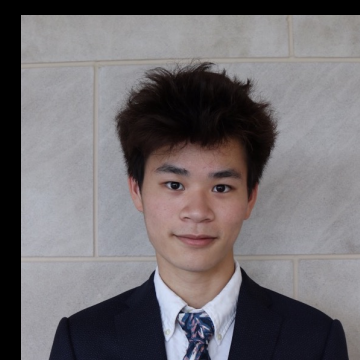
Kamilė Stankevičiūtė*



Anmol Kabra*



Raphael Thesmar



Johann Lee



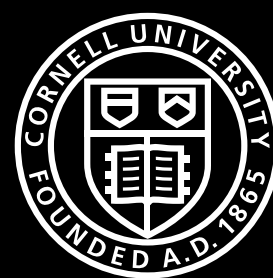
Julius Klenke



Carla P. Gomes



Kilian Q. Weinberger



Cornell University



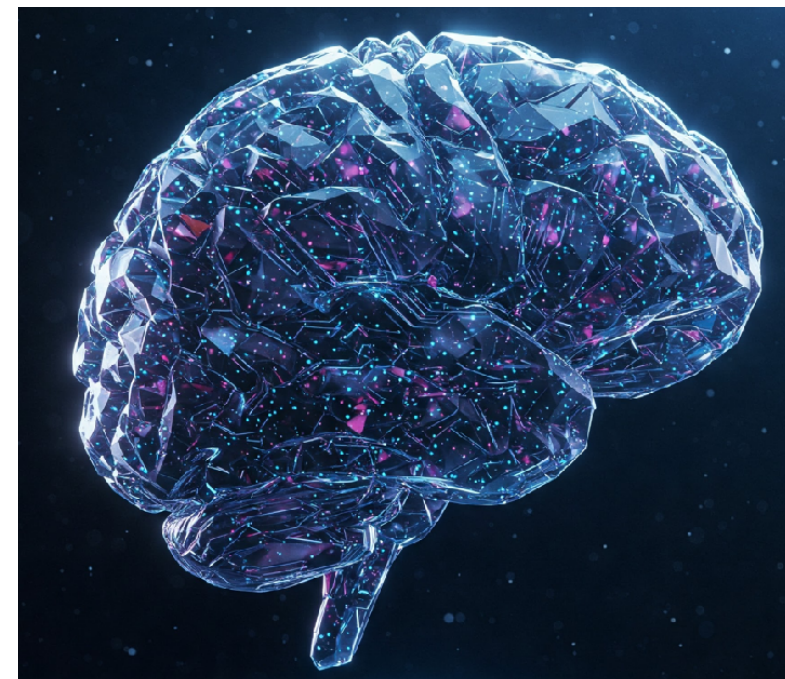
UNIVERSITY OF
CAMBRIDGE

*Equal contribution

What LLMs learn

Language Competency

Factual Knowledge



Model weights

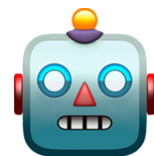
Common Knowledge



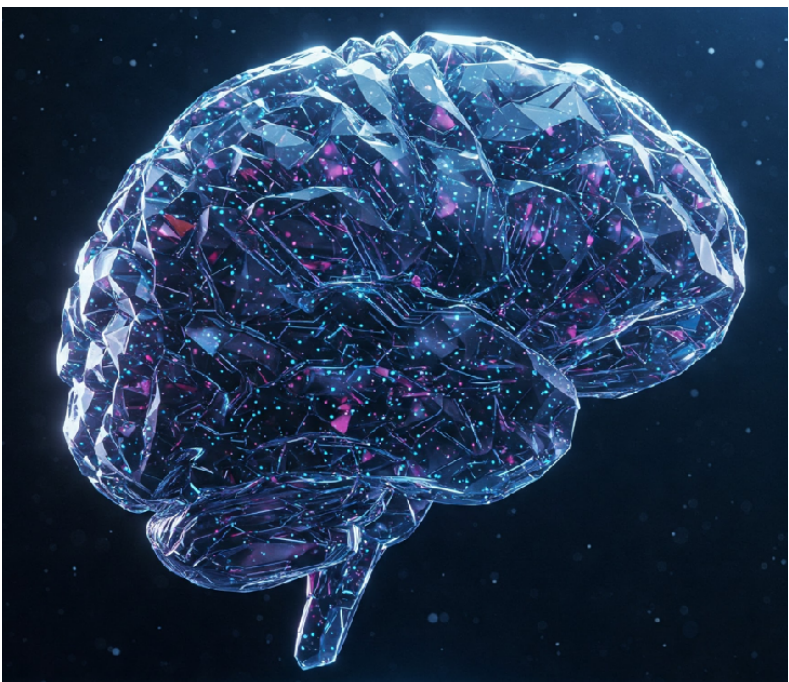
When was Napoleon born?

Language Competency

Napoleon was born on 8/15/1769.



Factual Knowledge



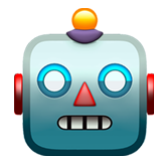
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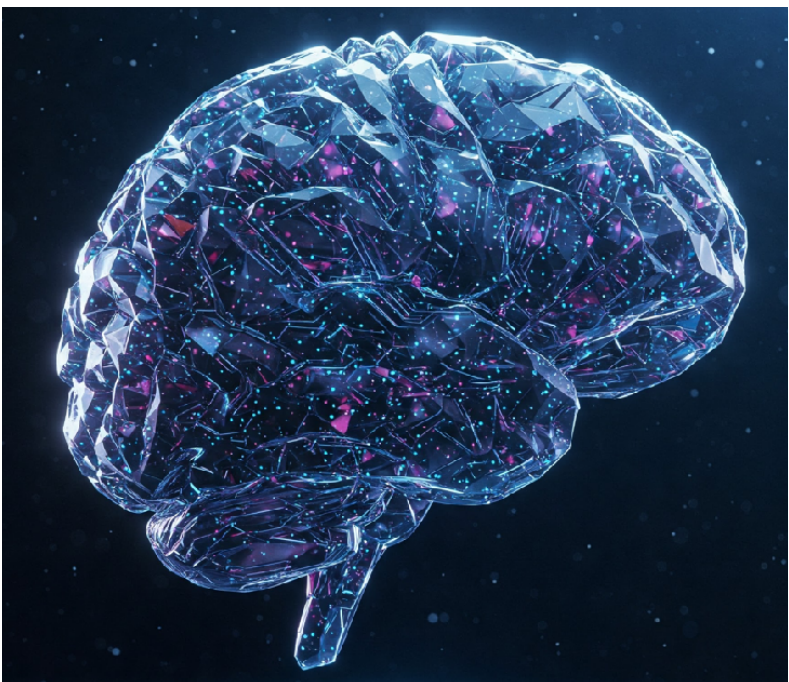
Language Competency

Subject-Verb-Object

Pronoun-Antecedent
Agreement

Proper Punctuation

Factual Knowledge



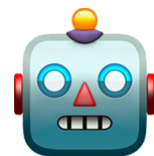
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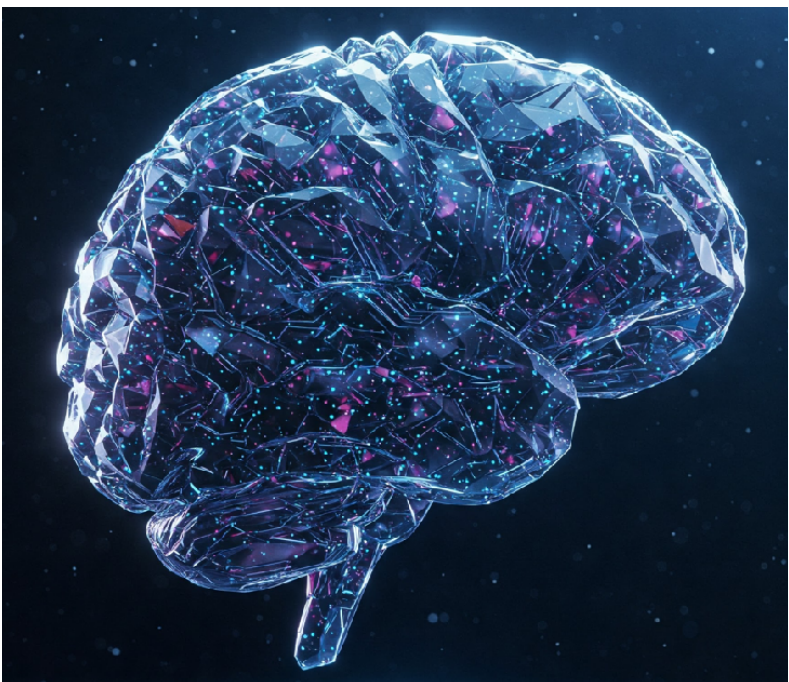
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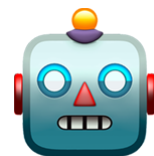
Napoleon was born

Napoleon had a mother



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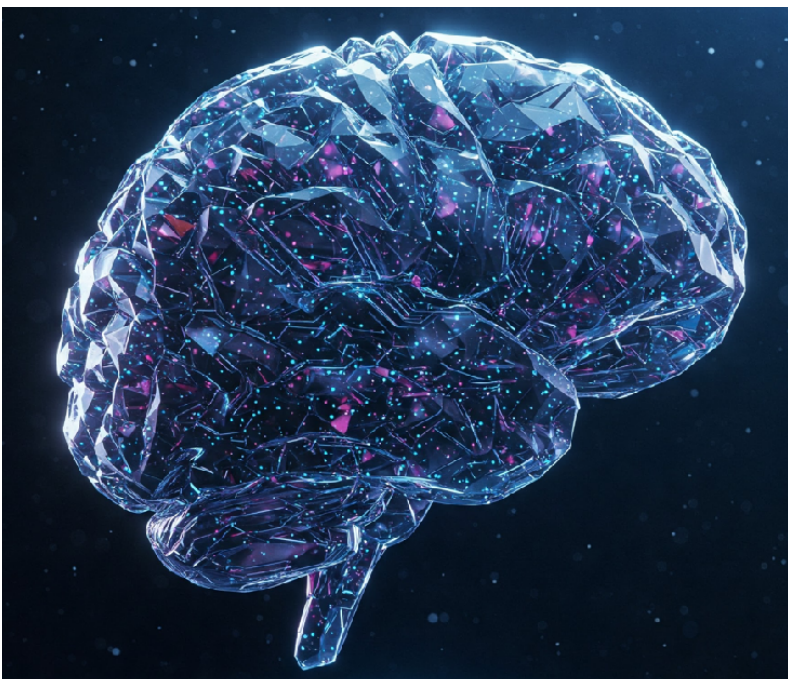


Language Competency

Subject-Verb-Object

Pronoun-Antecedent Agreement

Proper Punctuation



Model weights

Factual Knowledge

Napoleon was born on 8/15/1769

Napoleon’s mother was Letizia Bonaparte

Napoleon’s birthplace was Ajaccio, Corsica

Common Knowledge

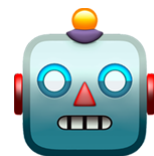
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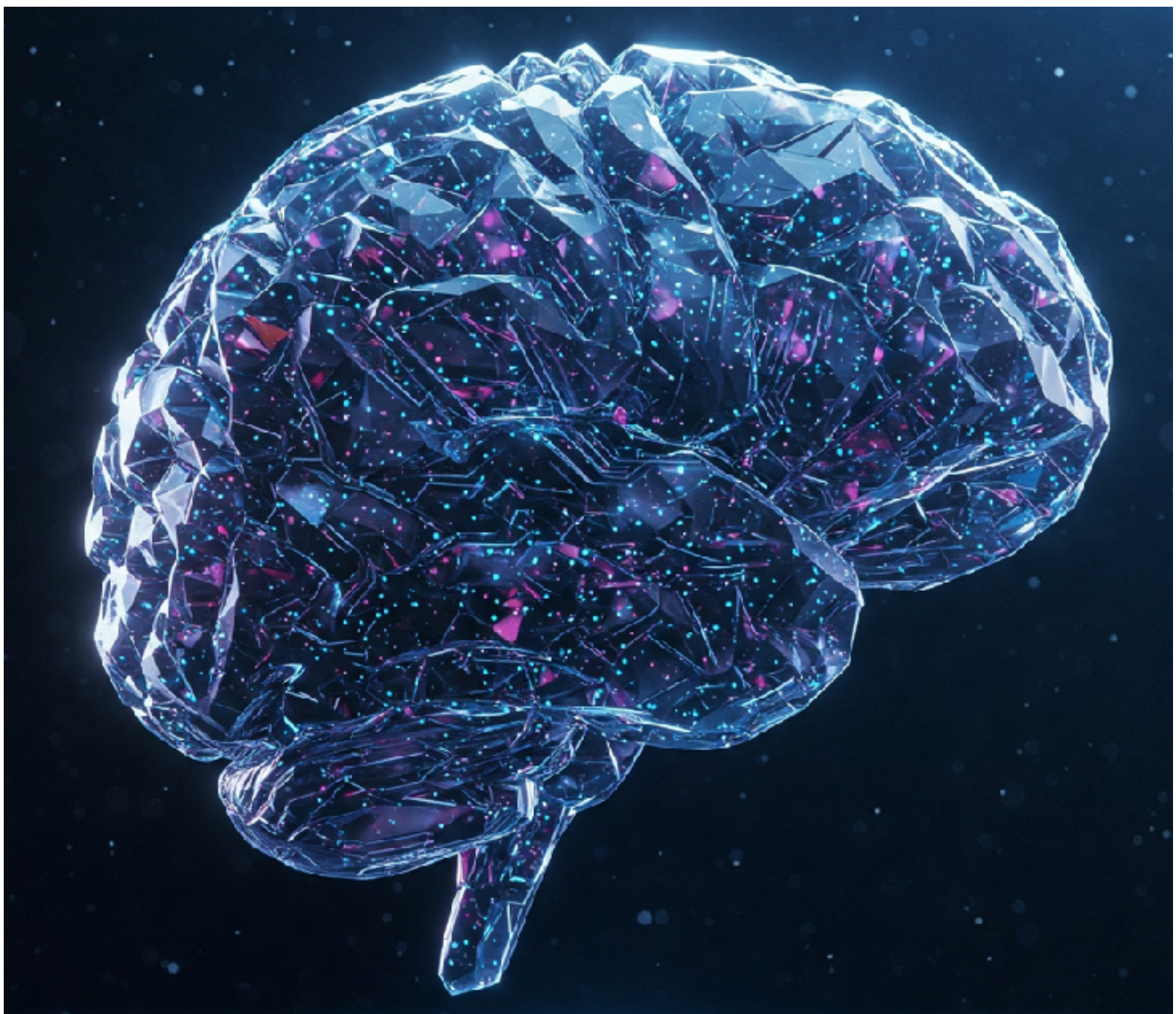


Language Competency

Subject-Verb-Object

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Proper Punctuation



Common Knowledge

Napoleon was born

Napoleon had a mother

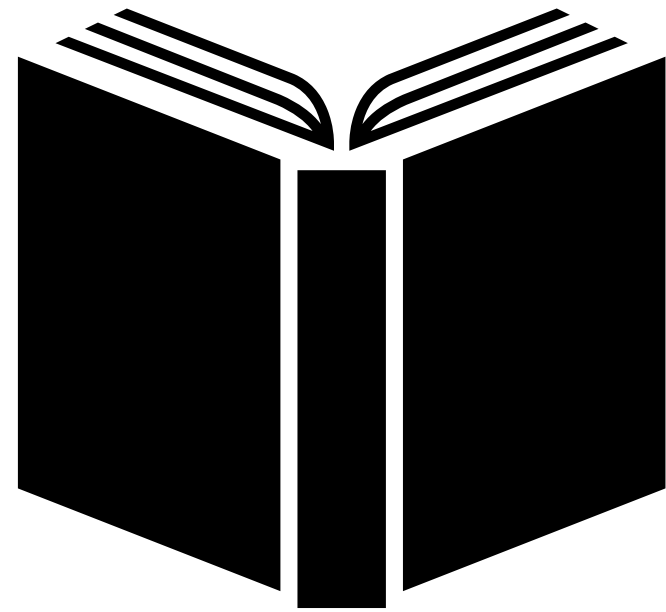
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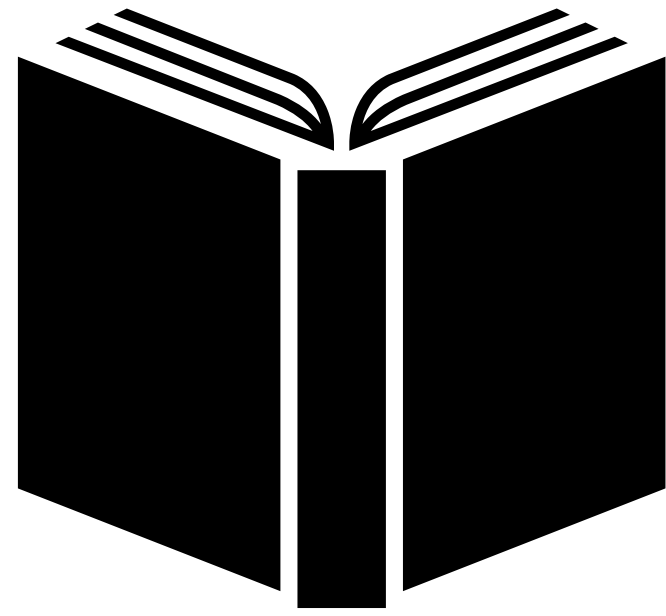
Externalizing Factual Knowledge



In-context

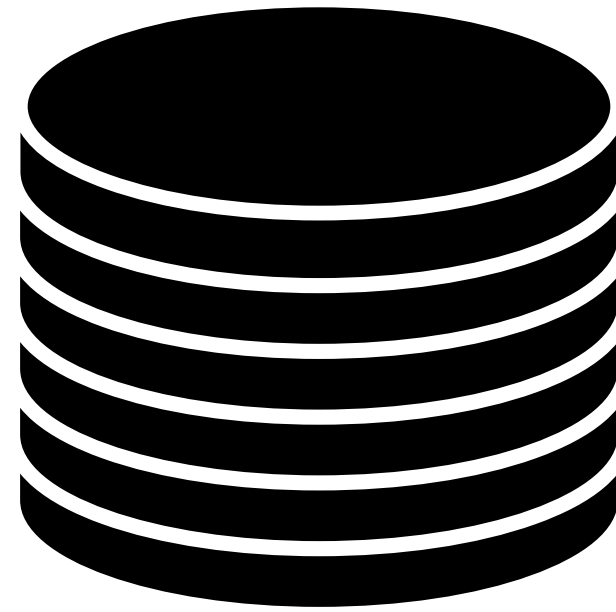
Claude 2, GPT 4
Turbo, Gemini 1.5,
Llama 3

Externalizing Factual Knowledge



In-context

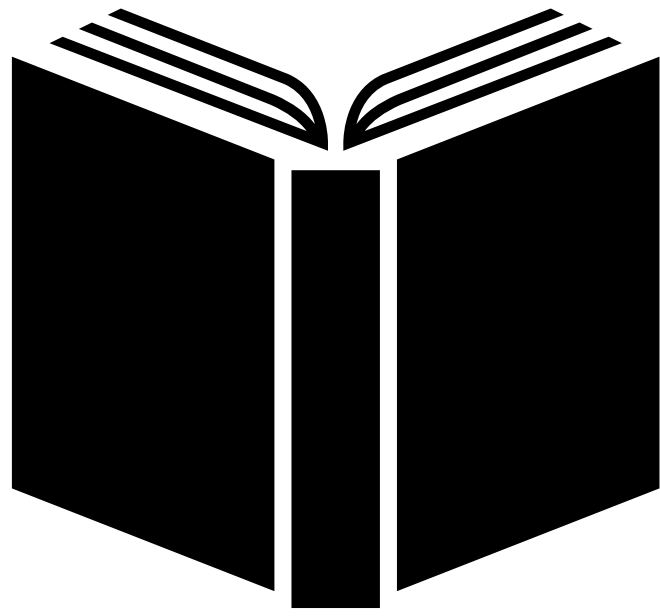
Claude 2, GPT 4
Turbo, Gemini 1.5,
Llama 3



RAG

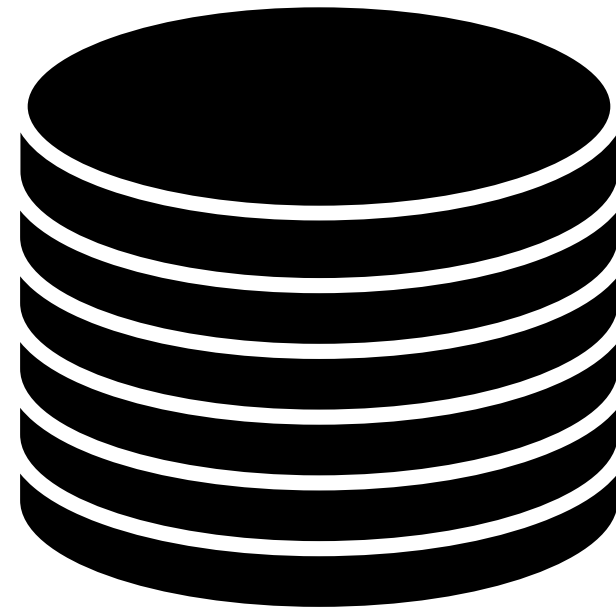
Chen et al. 2017,
Lewis et al. 2020

Externalizing Factual Knowledge



In-context

Claude 2, GPT 4
Turbo, Gemini 1.5,
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RAG

Chen et al. 2017,
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Agentic

Toolformer, ReAct,
Reflexion

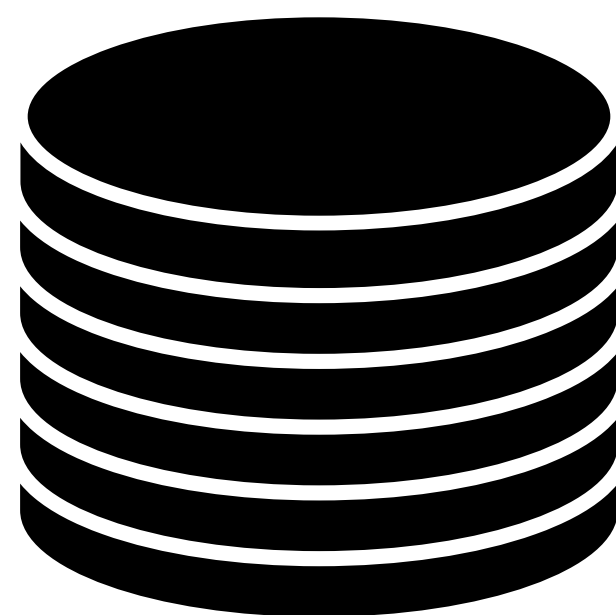
Externalizing Factual Knowledge



In-context

Claude 2, GPT 4
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Llama 3

NIAH, LongBench,
 ∞ -Bench, RULER,
HELMET



RAG

Chen et al. 2017,
Lewis et al. 2020

HotpotQA,
2WikiMultiHopQA,
MuSiQue



Agentic

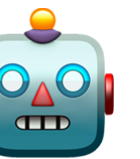
Toolformer, ReAct,
Reflexion

Gorilla,
ToolBench,
ToolQA

“Failure” Case from 2WikiMultiHopQA



Who is Rostislav I Of Kiev's paternal grandfather?



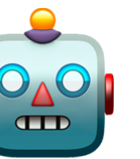
*Llama-3.3-70B + BM25

“Failure” Case from 2WikiMultiHopQA



Who is Rostislav I Of Kiev's paternal grandfather?

Based on the evidence, Rostislav I of Kiev is the son of
Mstislav I of Kiev.



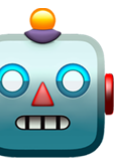
*Llama-3.3-70B + BM25

“Failure” Case from 2WikiMultiHopQA



Who is Rostislav I Of Kiev's paternal grandfather?

Based on the evidence, Rostislav I of Kiev is the son of Mstislav I of Kiev.
Also, Mstislav I of Kiev is the son of Vladimir II Monomakh,
but the evidence does not explicitly state this.



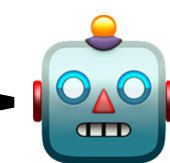
*Llama-3.3-70B + BM25

“Failure” Case from 2WikiMultiHopQA



Who is Rostislav I Of Kiev's paternal grandfather?

Based on the evidence, Rostislav I of Kiev is the son of Mstislav I of Kiev.
Also, Mstislav I of Kiev is the son of Vladimir II Monomakh, **but the evidence does not explicitly state this.**
However, according to historical records, Mstislav I of Kiev's father is indeed Vladimir II Monomakh. Therefore, Rostislav I of Kiev's paternal grandfather is Vladimir II Monomakh. The answer is **Vladimir II Monomakh**.



*Llama-3.3-70B + BM25

Desiderata for Synthetic Question-Answering Dataset

Corpus

- Model has no prior factual knowledge
- Scales to millions of articles
- Cheap to generate (bonus: on CPU or limited GPUs)

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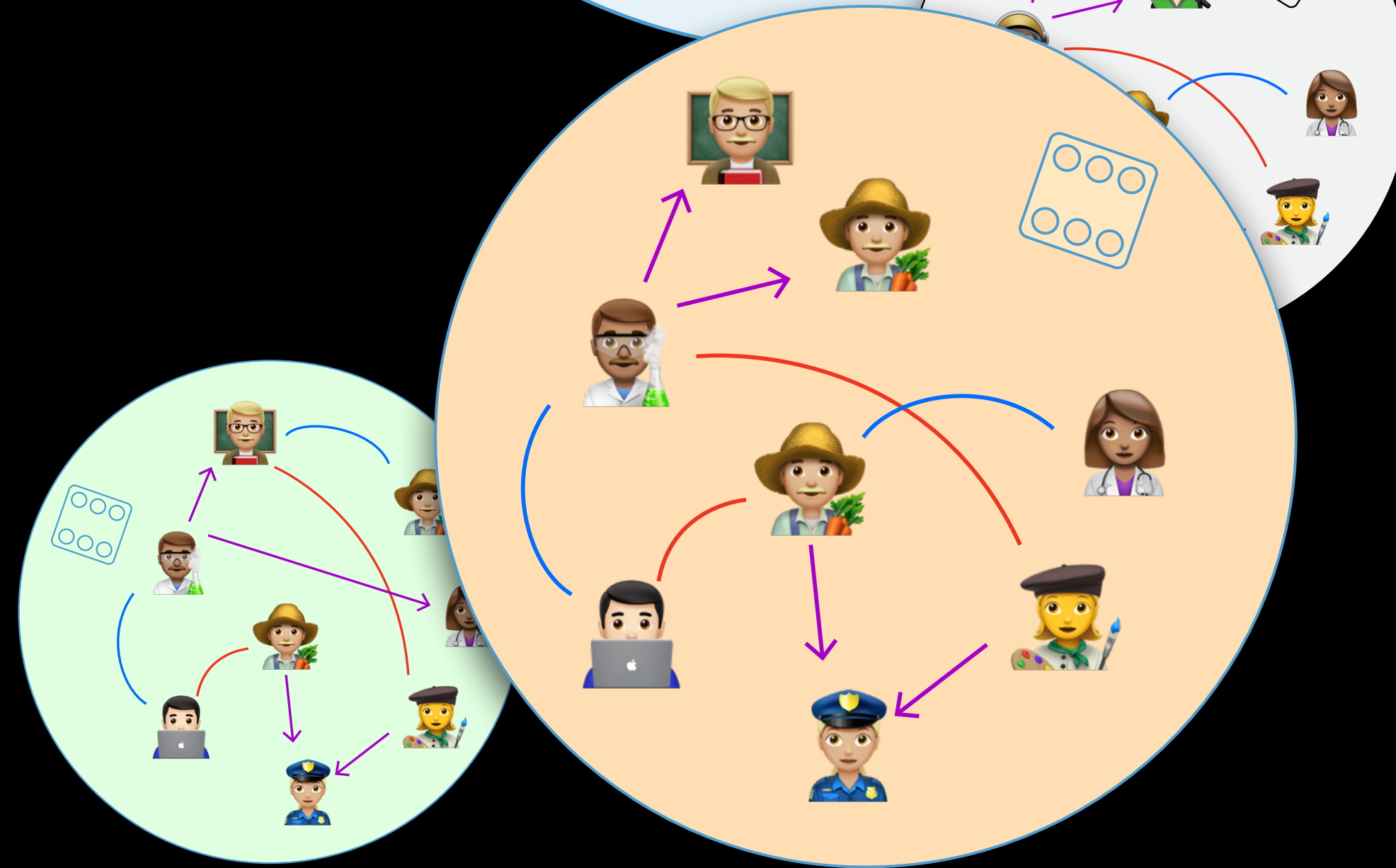
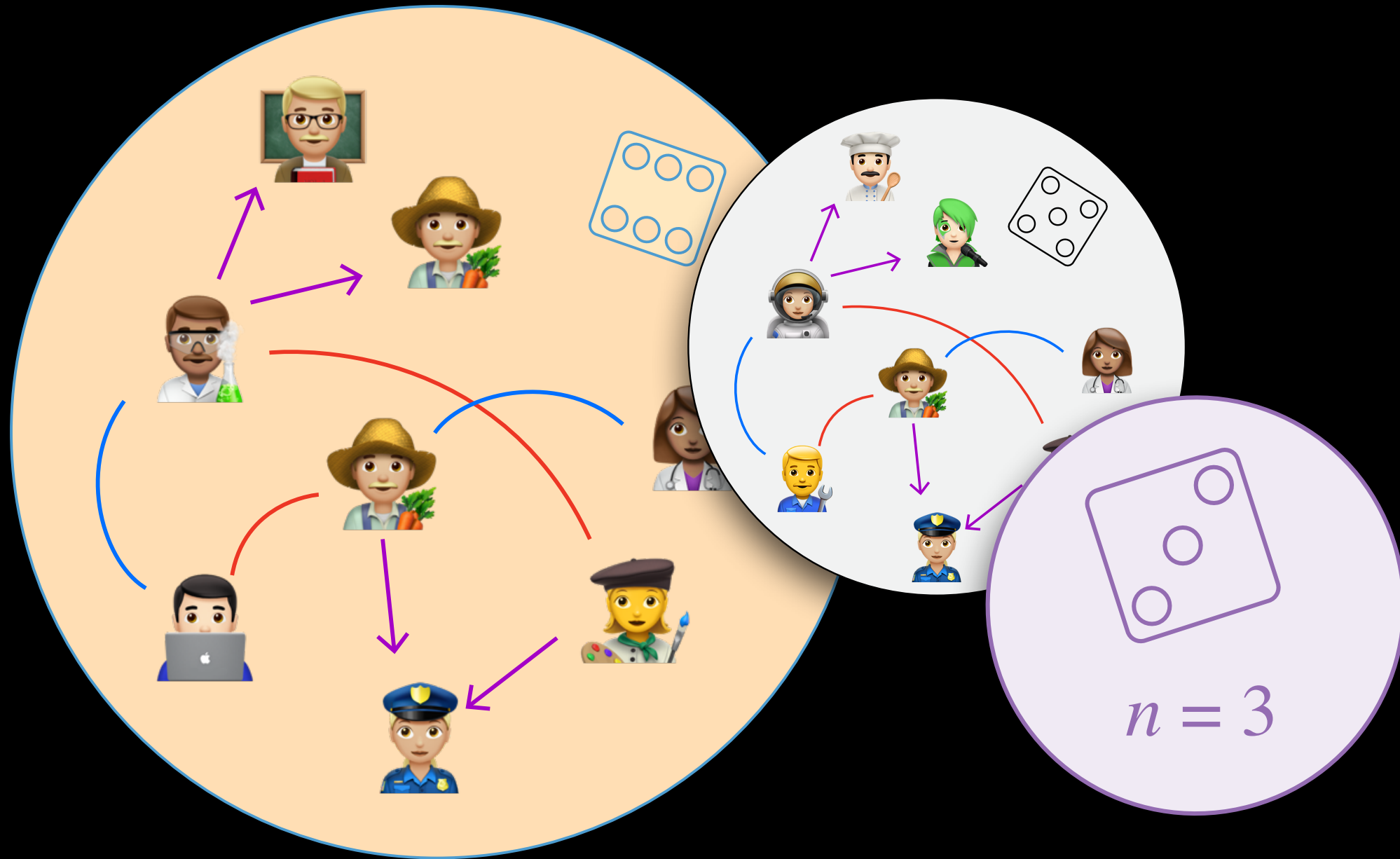
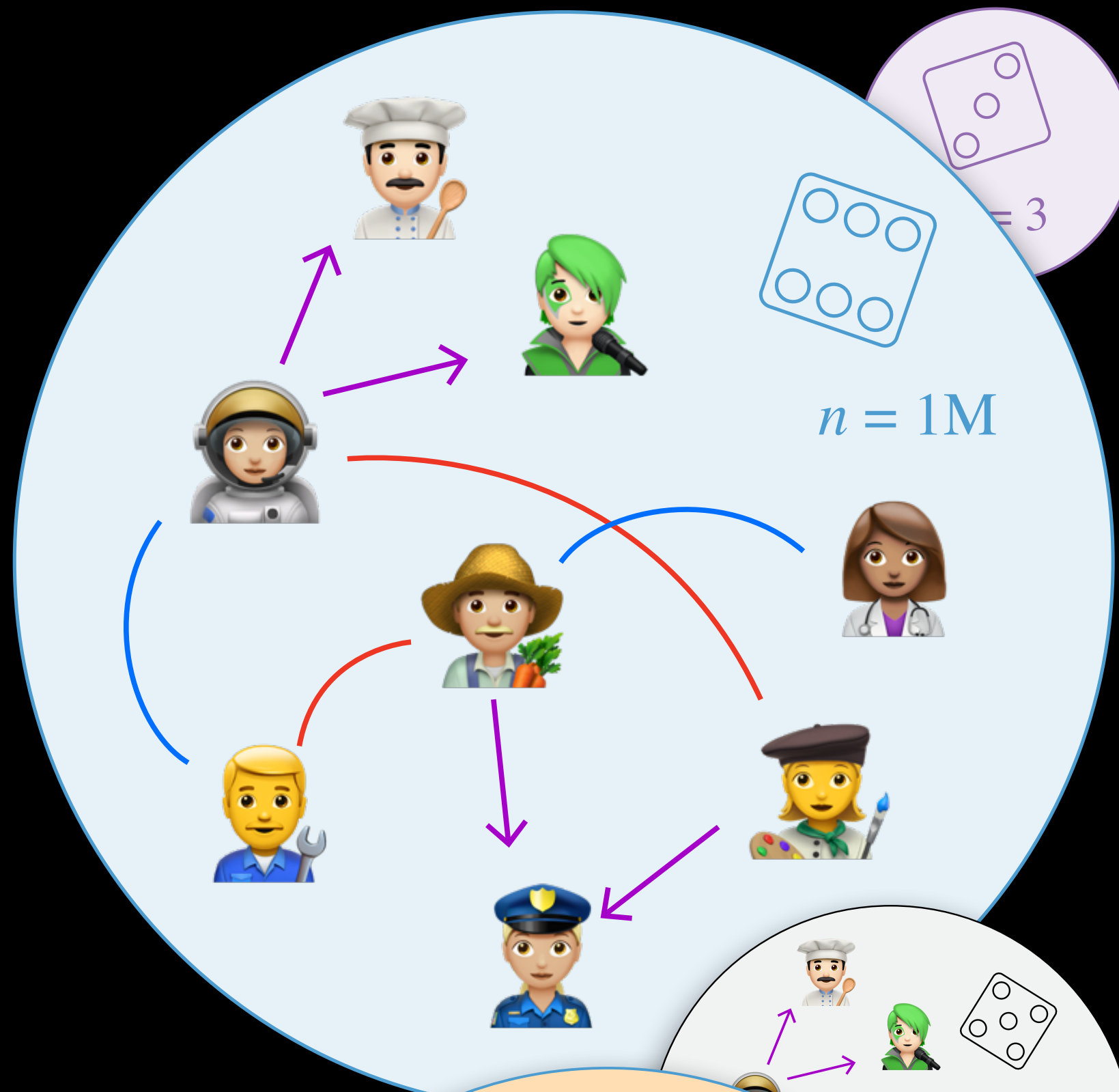
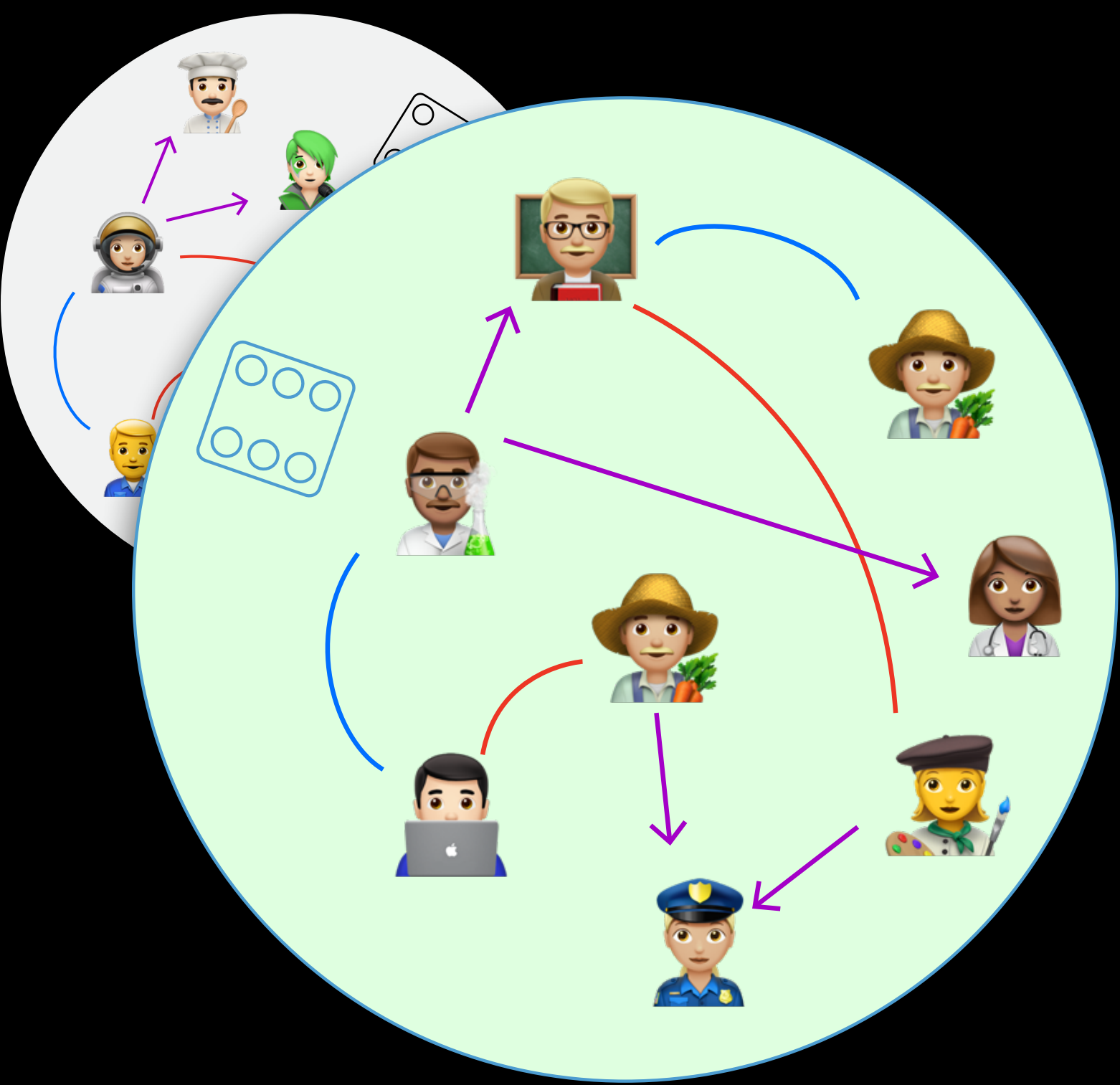
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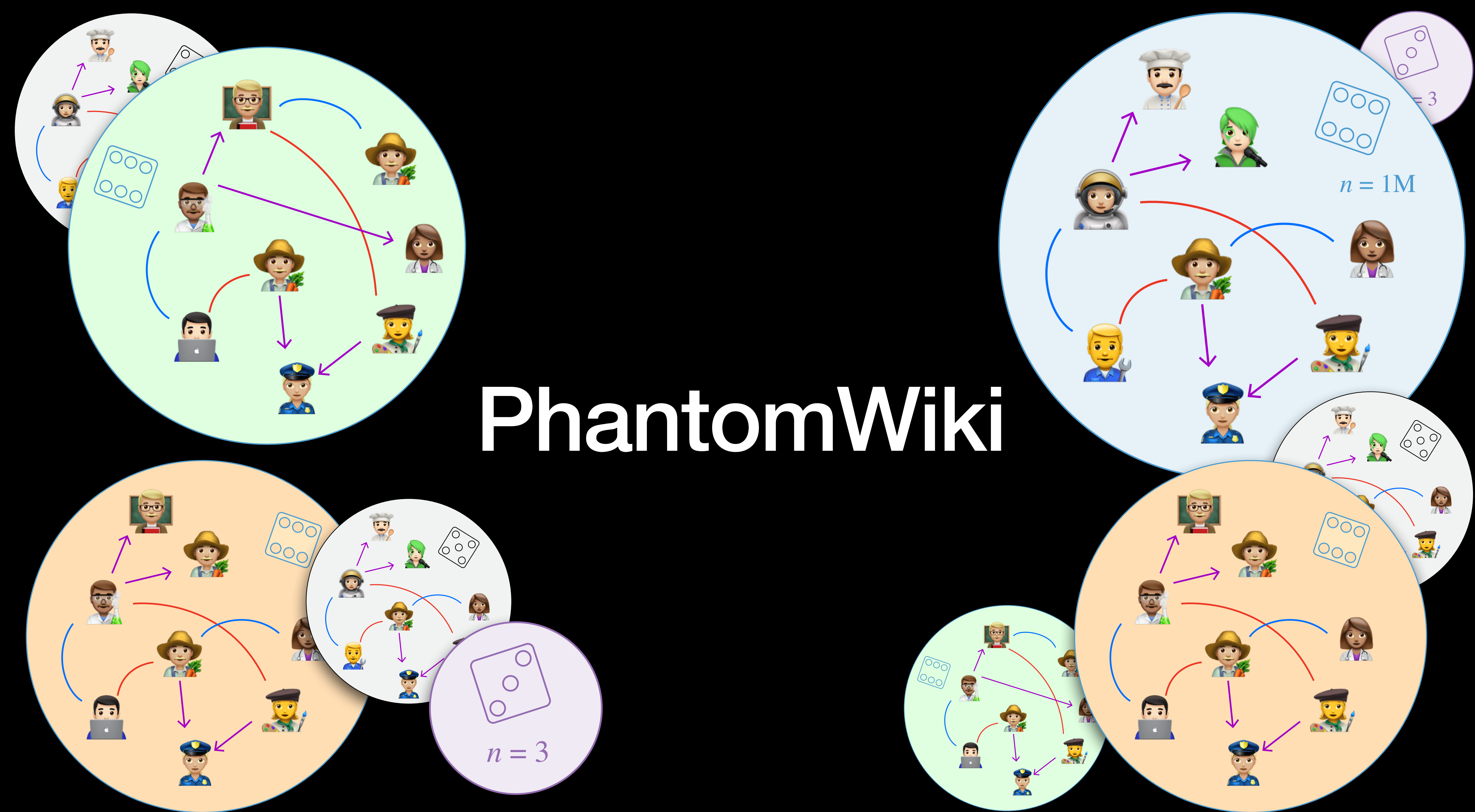
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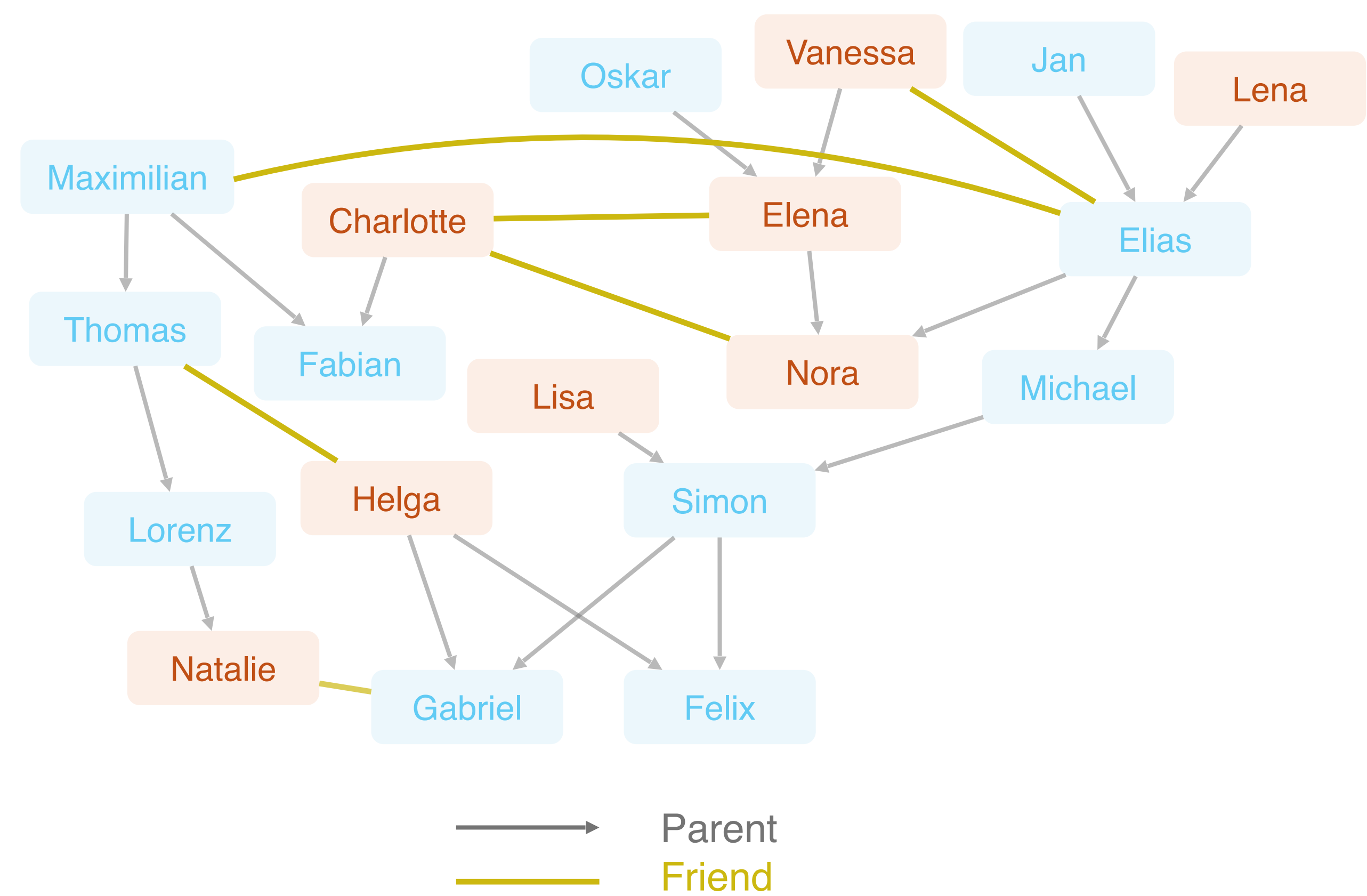
Impossible?



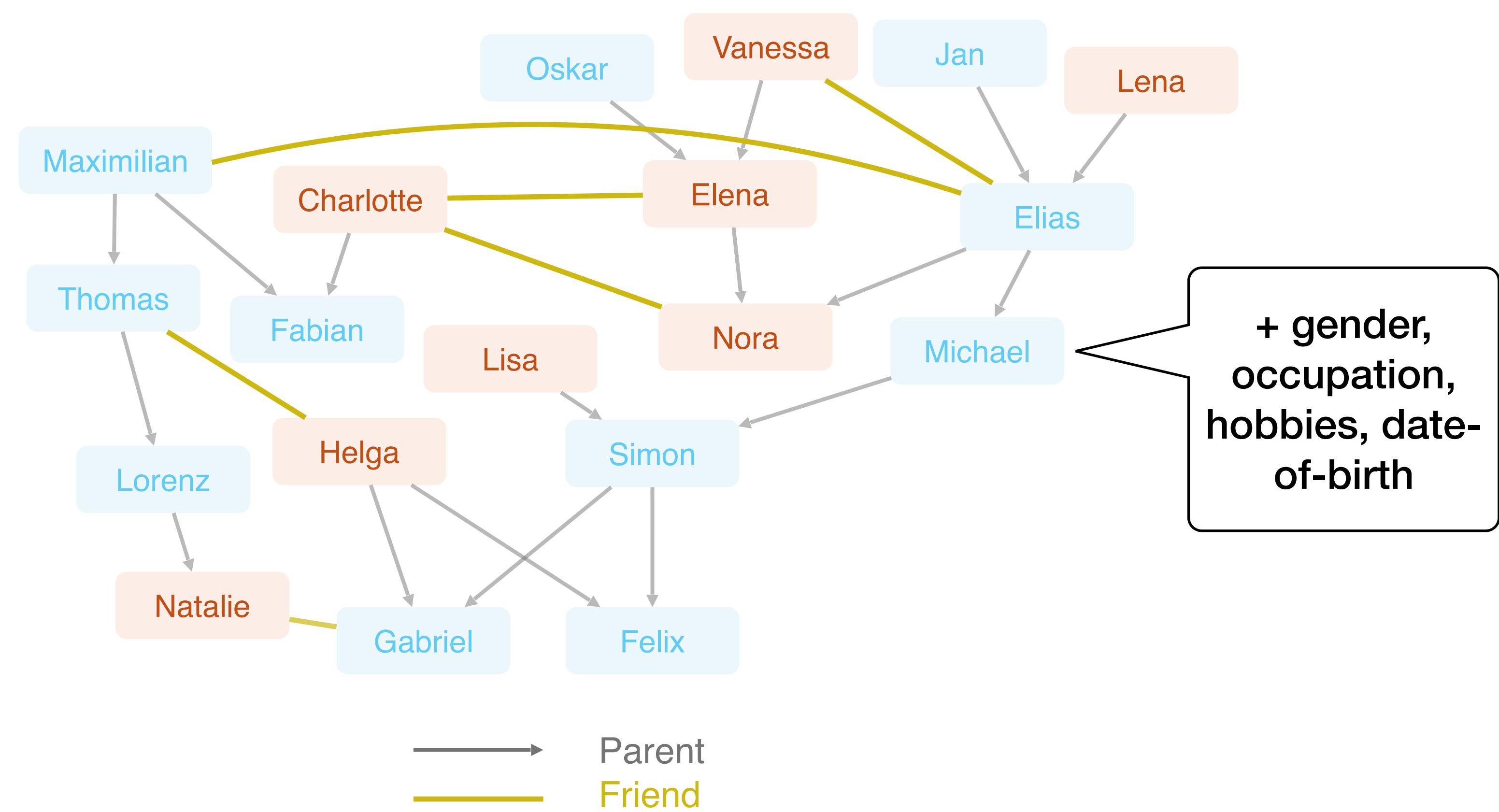
PhantomWiki



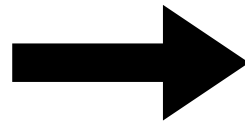
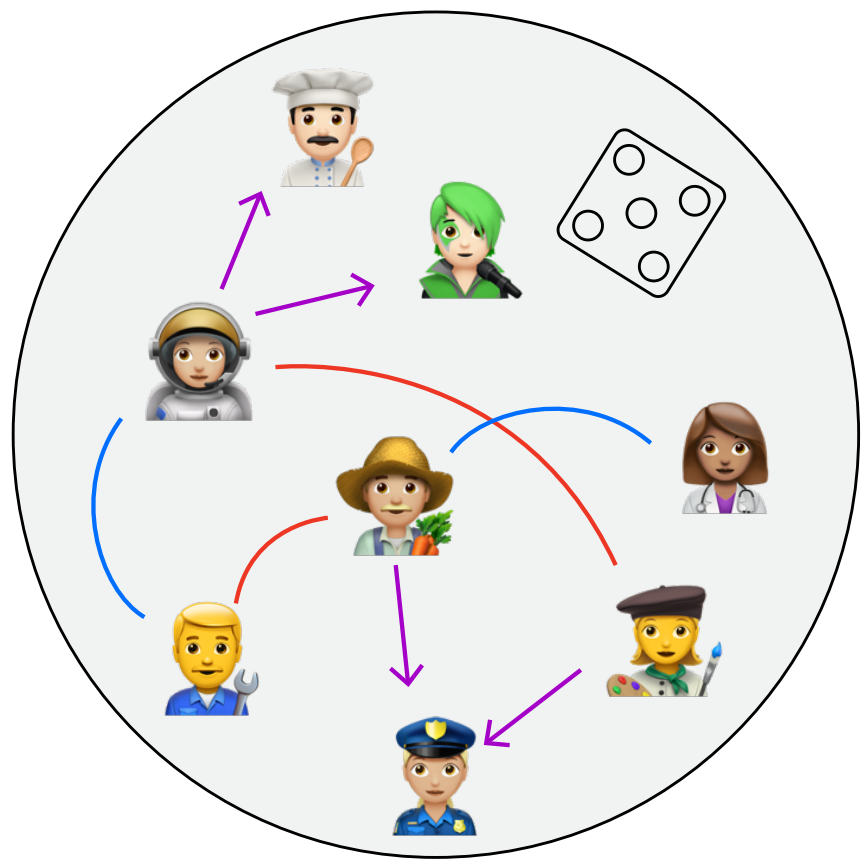
Step 1: Generating a Universe



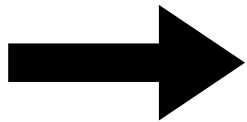
Step 1: Generating a Universe



Step 2: Universe → Corpus



?



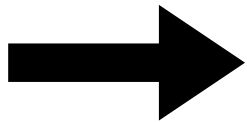
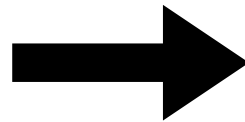
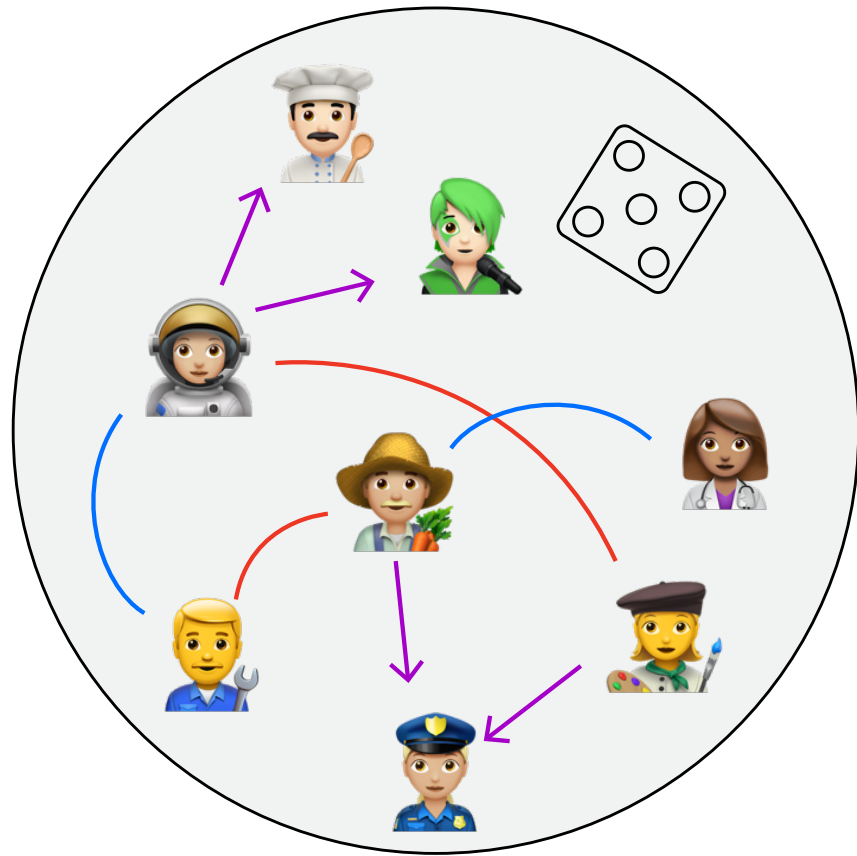
Alison Smock

Family
The daughters of Alison Smock are Cythia Smock, Shelli Beltran, Stacia Toombs.
The husband of Alison Smock is Williams Smock.

Friends
The friends of Alison Smock are Courtney Parmer, Jamison Baptiste, Wilbert Toombs.

Attributes
The date of birth of Alison Smock is 0929-10-28.
The occupation of Alison Smock is broadcast engineer.
The hobby of Alison Smock is meteorology.
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Step 2: Universe → Corpus



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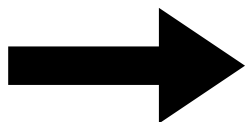
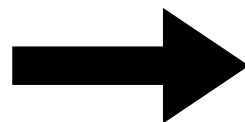
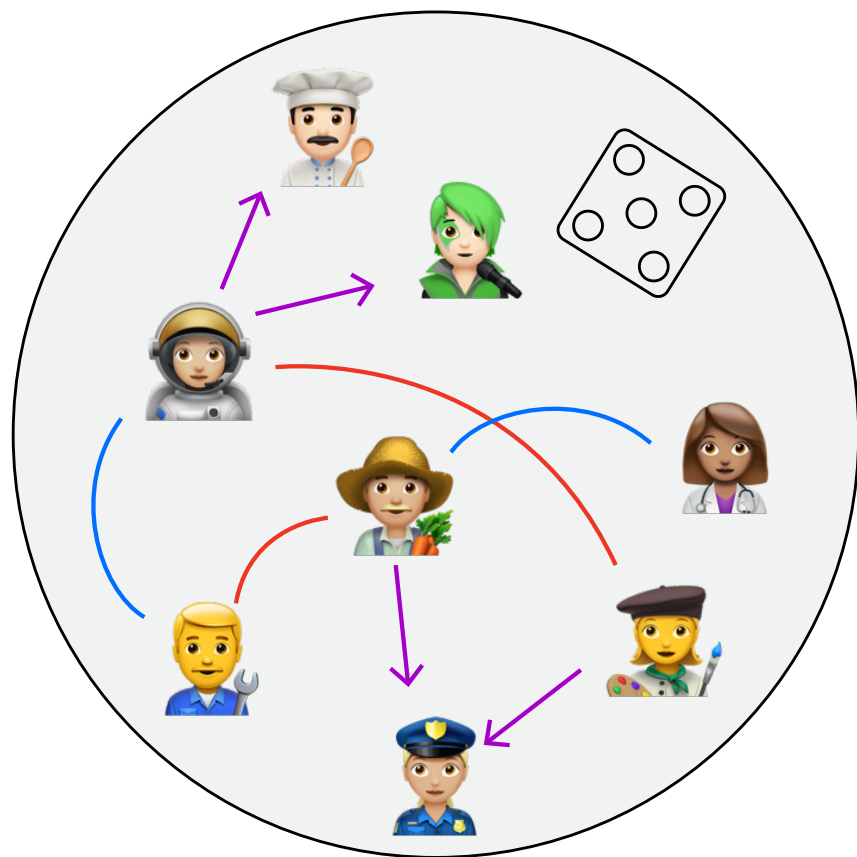
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Fact = predicate + constants

The husband of Alison Smock is Williams Smock.

husband(“Alison Smock”, “Williams Smock”).

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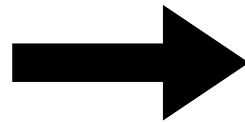
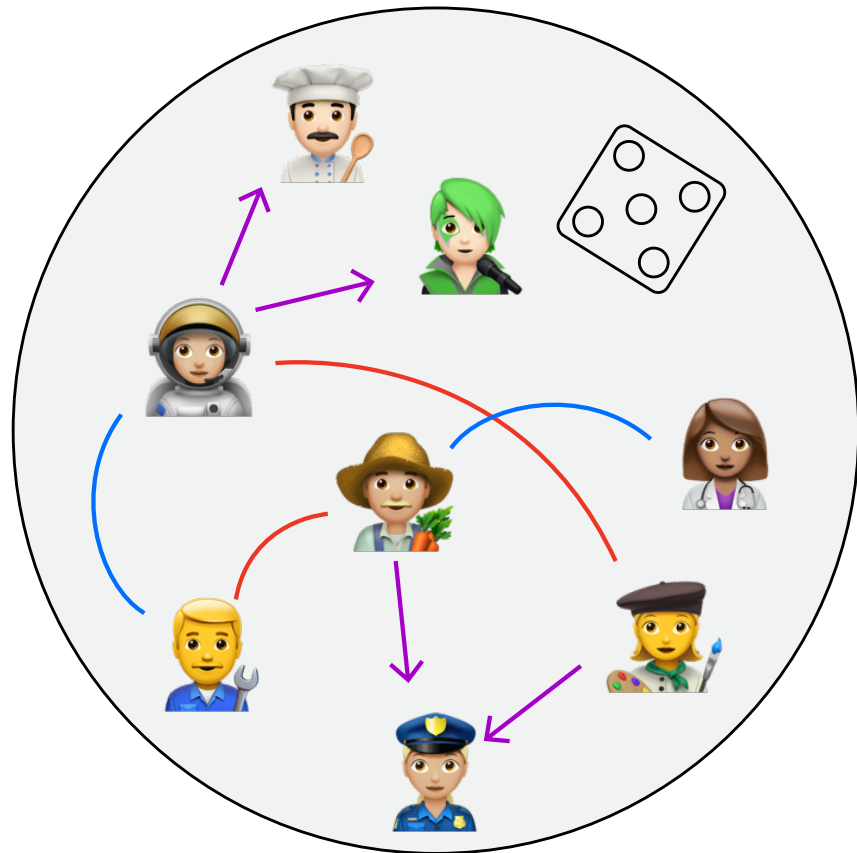


Prolog



Alain Colmerauer

Step 2: Universe → Corpus



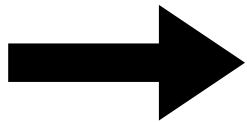
Prolog Database

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Prolog

Alain Colmerauer



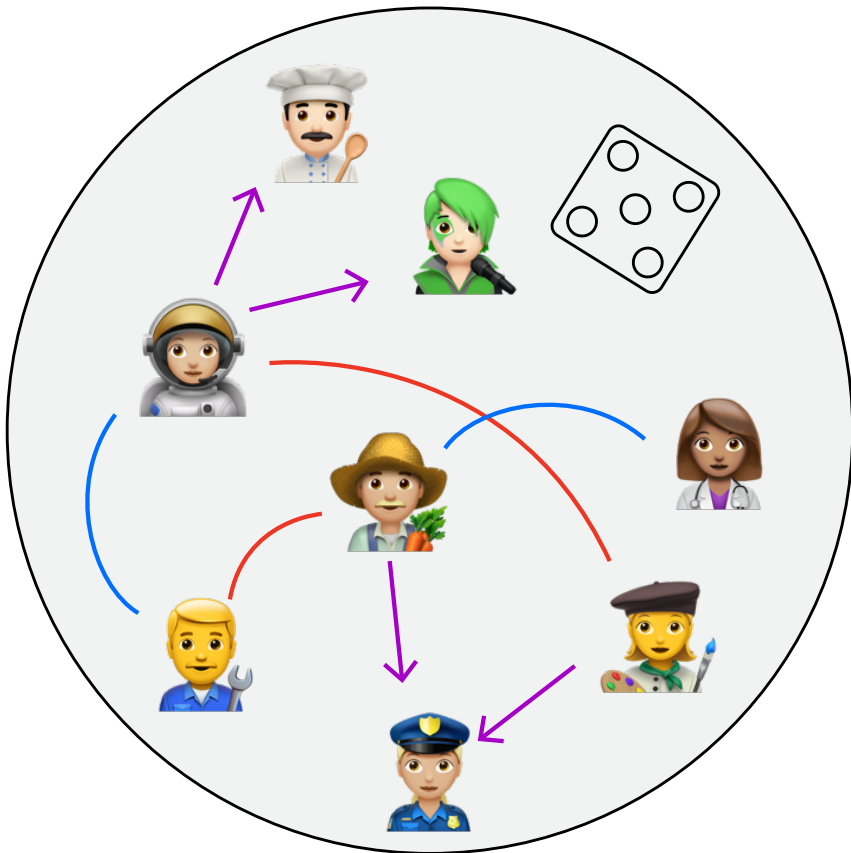
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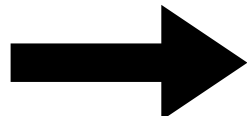
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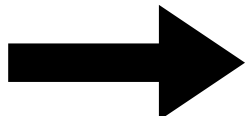
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Prolog Database



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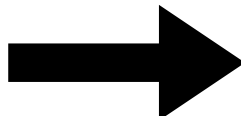
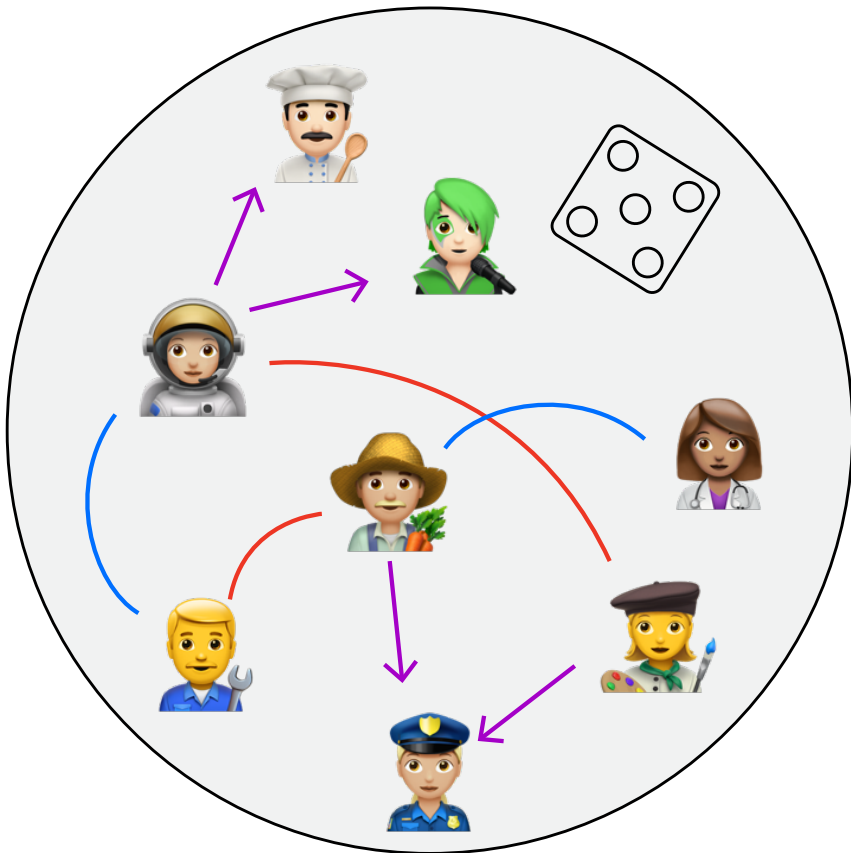
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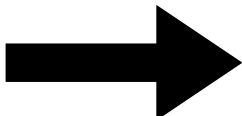
Prolog

Alain Colmerauer

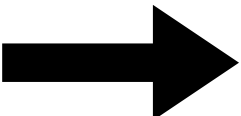
Step 2: Universe → Corpus



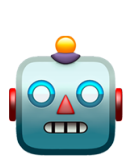
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(Optional)

Fact = predicate + constants

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husband(“Alison Smock”, “Williams Smock”).



Prolog



Alain Colmerauer

Alison Smock
Alison Smock, a brilliant and charismatic broadcast engineer, has been fascinated by the intricacies of technology and the wonders of the atmosphere for as long as anyone can remember. \red{Born on October 28, 1929}, Alison grew up with an insatiable curiosity and a passion for learning that would serve her well throughout her life. As a young girl, she was always tinkering with gadgets and machinery, trying to understand how they worked, and this innate curiosity eventually led her to pursue a career in broadcast engineering.

As she navigated her professional path, Alison met her husband, Williams Smock, and together they built a loving family. They had three beautiful daughters, Cythia, Shelli, and Stacia, who brought immense joy and fulfillment to their lives. According to her daughter Cythia, “Mom was always there for us, supporting us and encouraging us to pursue our dreams, no matter how big or small they seemed.” Alison’s family was her rock, and she cherished every moment she spent with them.

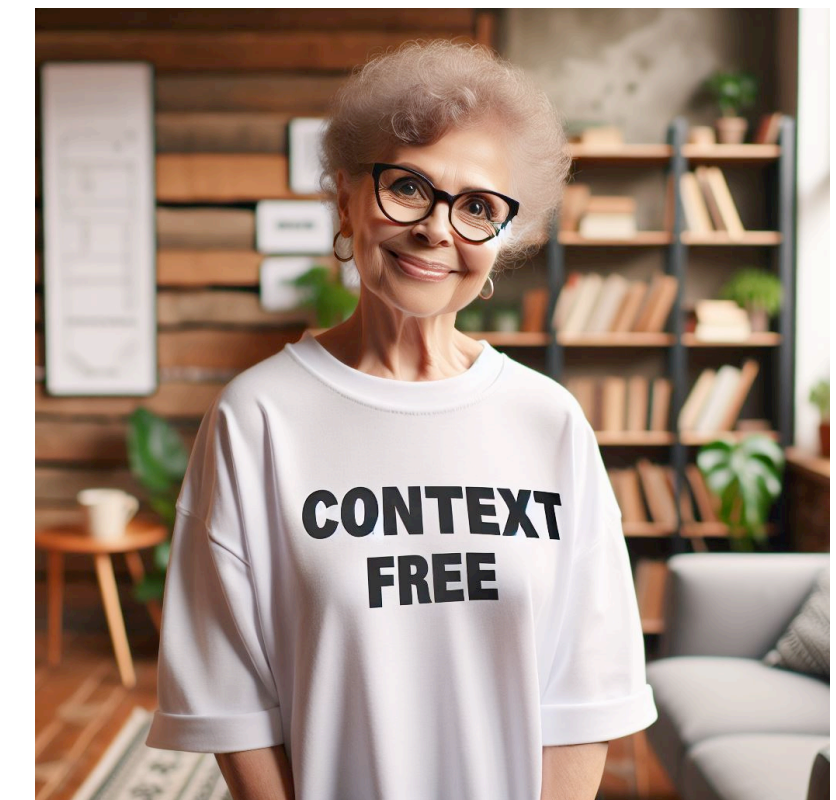
Alison’s career as a broadcast engineer was marked by numerous achievements and accomplishments. She was known for her exceptional problem-solving skills and her ability to stay calm under pressure. Her colleagues admired her dedication and work ethic, and she quickly became an invaluable member of her team. When she wasn’t working,

Step 3: Generating Questions



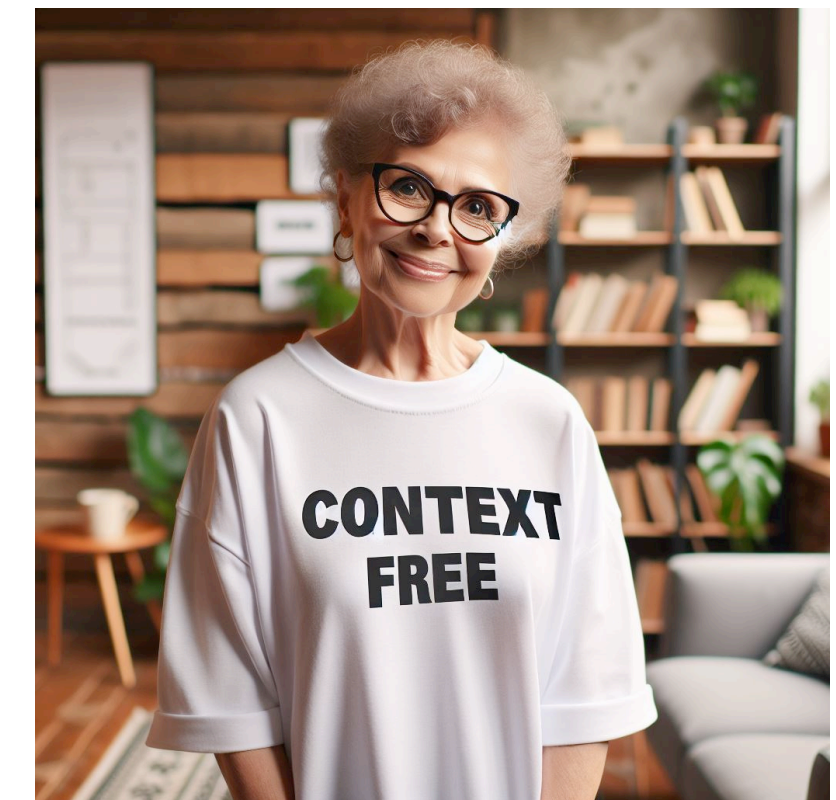
Step 3: Generating Questions

Who is <...> ?



Step 3: Generating Questions

Who is <...> ?
the ■ of <...>

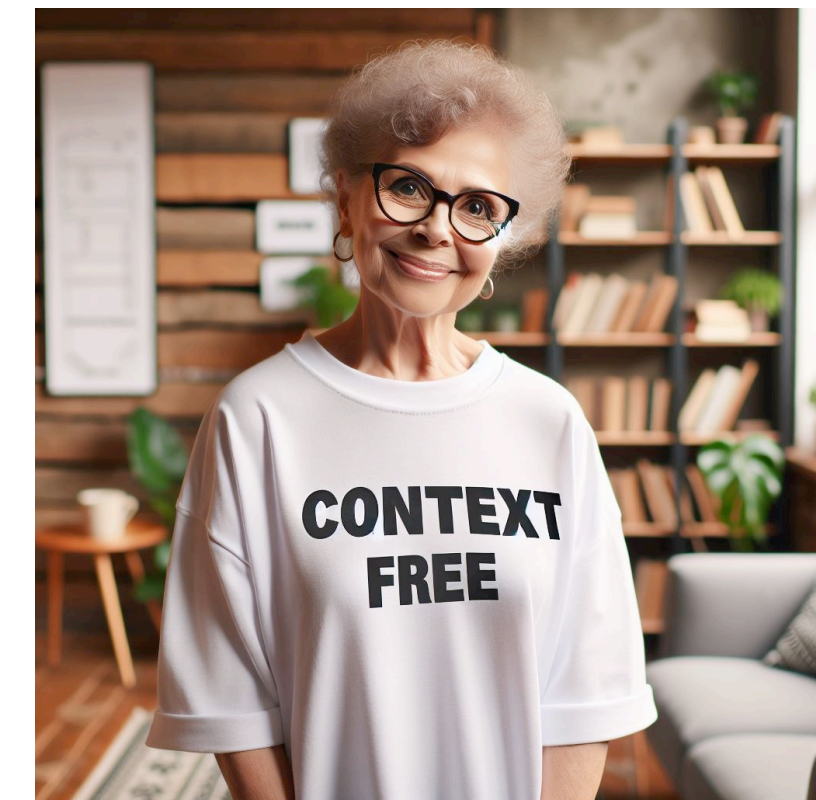


Step 3: Generating Questions

Who is <...> ?

the  of <...>

the  of <...>



Step 3: Generating Questions

Who is <...> ?

the  of <...>

the  of <...>

the person whose  is 



Step 3: Generating Questions

Who is <...> ?

the ■ of <...>

the ■ of <...>

the person whose ■ is ■

Template: Who is the ■ of the ■ of the person whose ■ is ■?



Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is .

hobby

running
reading
climbing
...

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

- | | |
|-------|---------------------------------------|
| hobby | running
reading
climbing
... |
| job | plumber
engineer
lawyer
... |

Step 3: Generating Questions

Template: Who is the of the of the person whose is ?

- hobby

running

reading

climbing

...
- job

plumber

engineer

lawyer

...

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is .

- mother
- father
- brother
- sister
- cousin
- uncle
- aunt
- friend
- ...

- | | |
|-------|----------|
| hobby | running |
| | reading |
| | climbing |
| | ... |
| job | plumber |
| | engineer |
| | lawyer |
| | ... |

Step 3: Generating Questions

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- brother
- sister
- cousin
- uncle
- aunt
- friend
- ...

- | | |
|---|---|
|  hobby |  reading |
| | climbing |
| | ... |
| job | plumber |
| | engineer |
| | lawyer |
| | ... |

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

- | | |
|---------|------------------|
| mother | mother |
| father | father |
| brother | brother |
| sister | sister |
| cousin | cousin |
| uncle | <div>uncle</div> |
| aunt | aunt |
| friend | friend |
| ... | ... |

- | | |
|------------------|--|
| <div>hobby</div> | <div>running</div> <div><div>reading</div></div> <div>climbing</div> |
| | ... |
| job | plumber
engineer
lawyer
... |

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

mother	mother	hobby	running
father	father		reading
brother	brother		climbing
sister	sister		...
cousin	cousin	job	plumber
uncle	uncle		engineer
aunt	aunt		lawyer
friend	friend		...
...	...		

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is .

mother	mother	hobby	running
father	father		reading
brother	brother		climbing
sister	sister		...
cousin	cousin	job	plumber
uncle	uncle		engineer
aunt	aunt		lawyer
friend	friend		...
...	...		

Question: “Who is the mother of the father of the person whose hobby is reading?”

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

mother	mother	hobby	running
father	father		reading
brother	brother		climbing
sister	sister		...
cousin	cousin	job	plumber
uncle	uncle		engineer
aunt	aunt		lawyer
friend	friend		...
...	...		

Question: “Who is the mother of the father of the person whose hobby is reading?”

Question difficulty $\approx$ Sum of predicate difficulties

Step 3: Generating Questions

Template: Who is the  of the  of the person whose  is ?

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Question: “Who is the mother of the father of the person whose hobby is reading?”

Question difficulty $\approx$ Sum of predicate difficulties

Step 4: Generating Ground-Truth Answers

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Step 4: Generating Ground-Truth Answers

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`hobby(X, “reading”), uncle(X, Y), mother(Y, Z)`

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Prolog Database

Step 4: Generating Ground-Truth Answers

Question: “Who is the mother of the father of the person whose hobby is reading?”

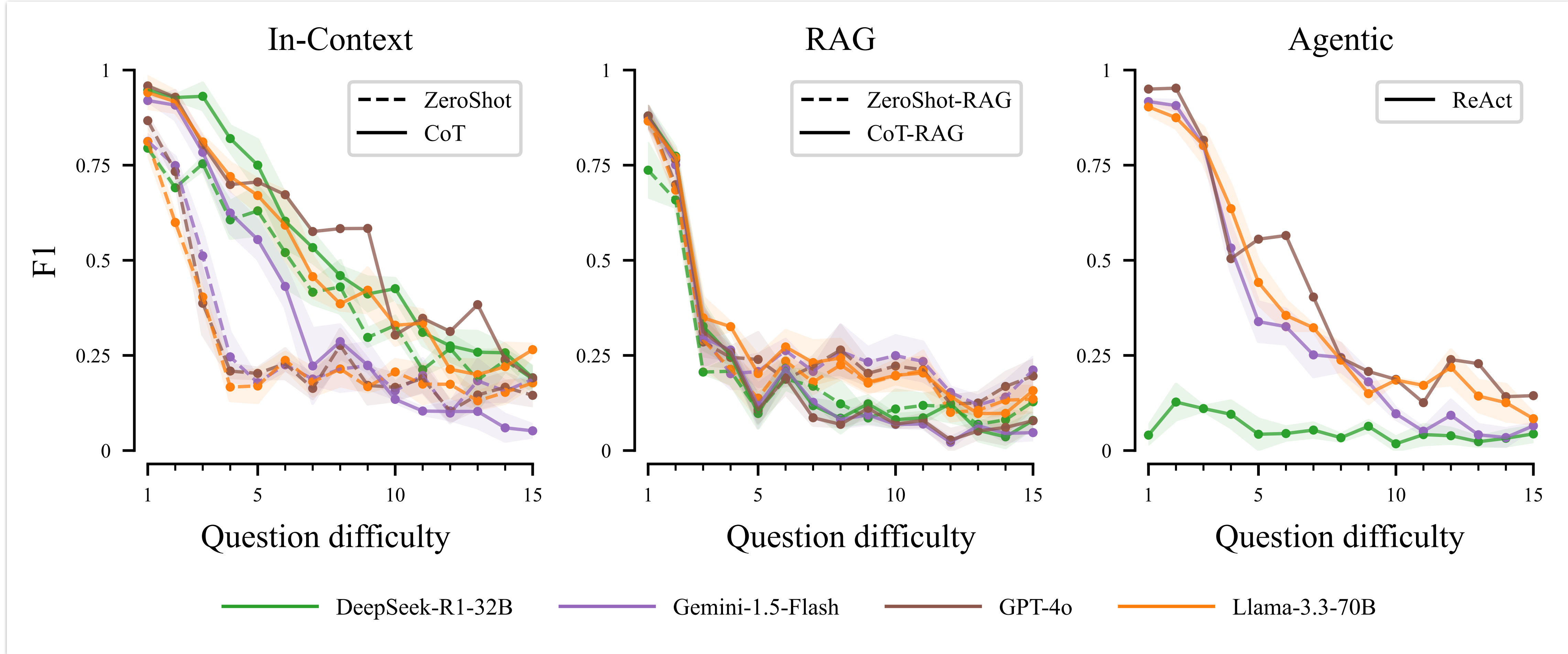
`hobby(X, “reading”), uncle(X, Y), mother(Y, Z)`



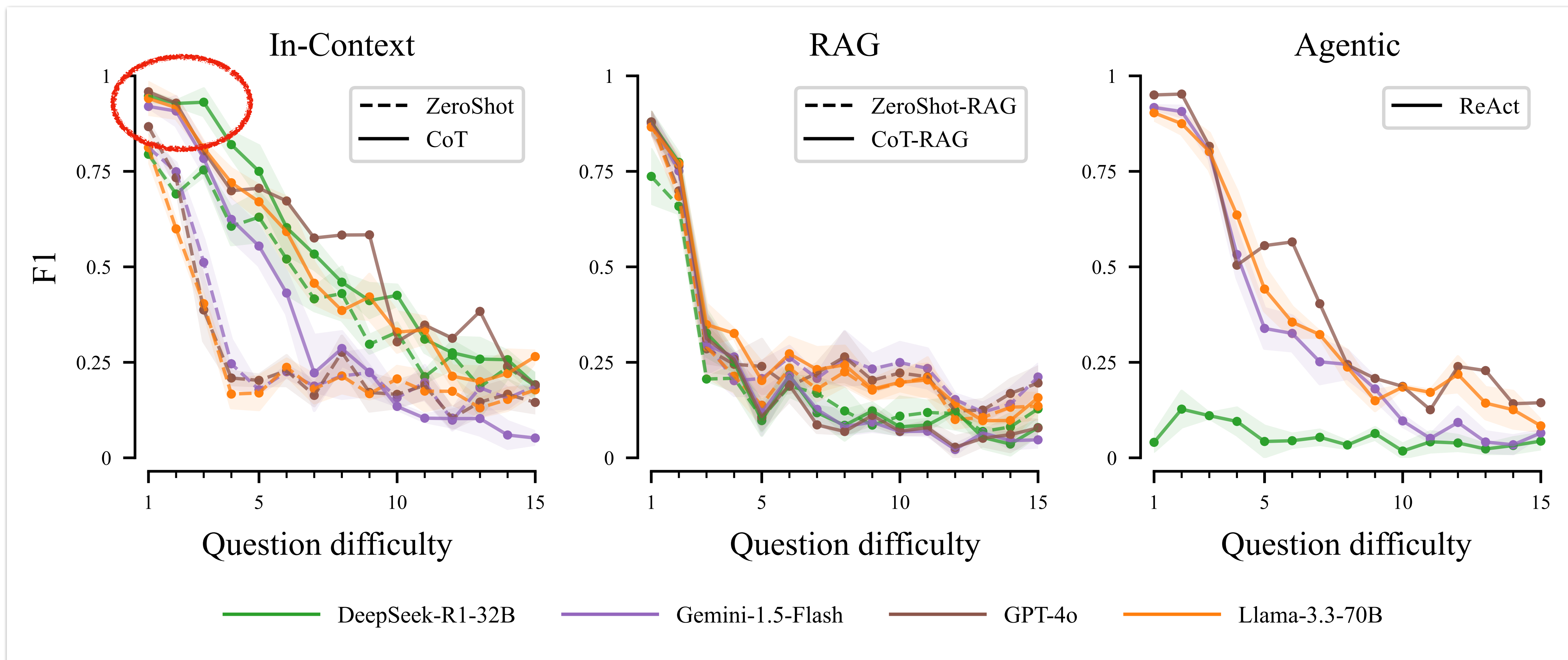
Prolog Database

Z ← Cythia Smock

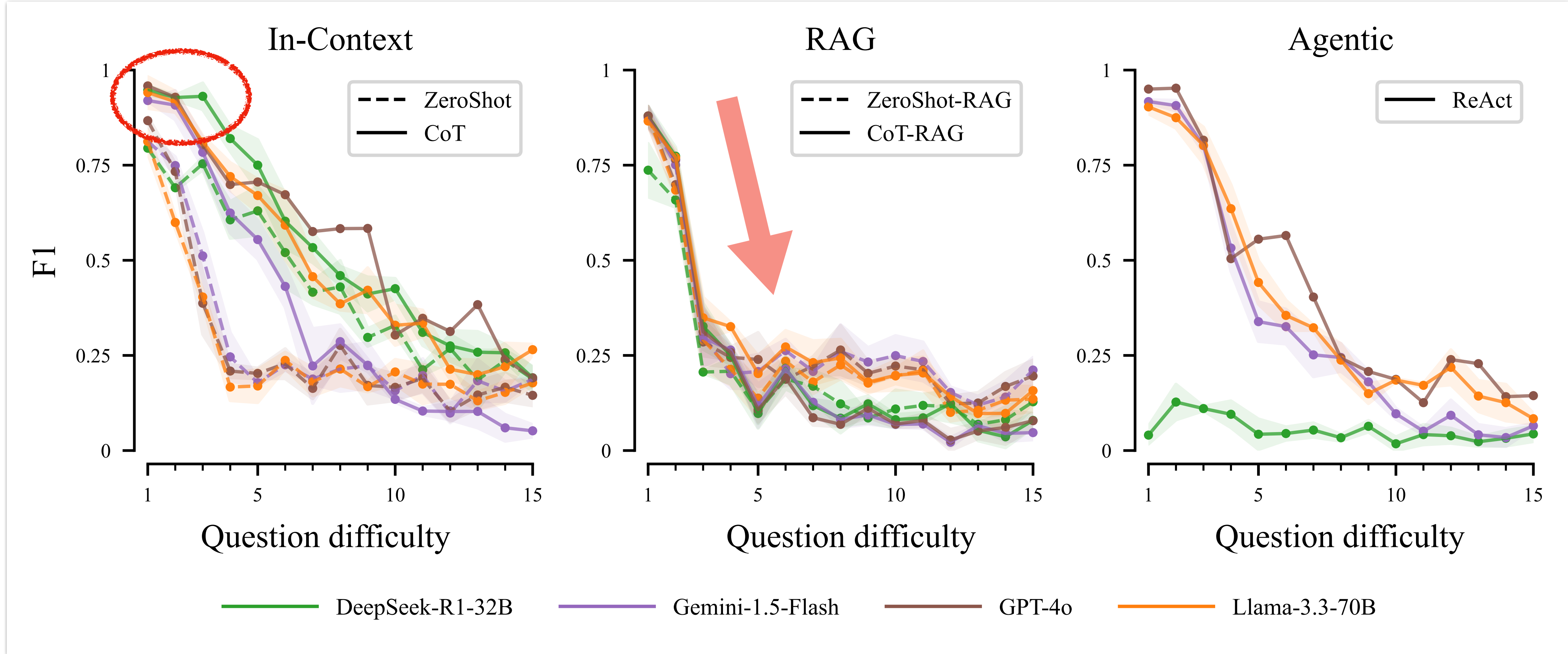
Evaluating Reasoning



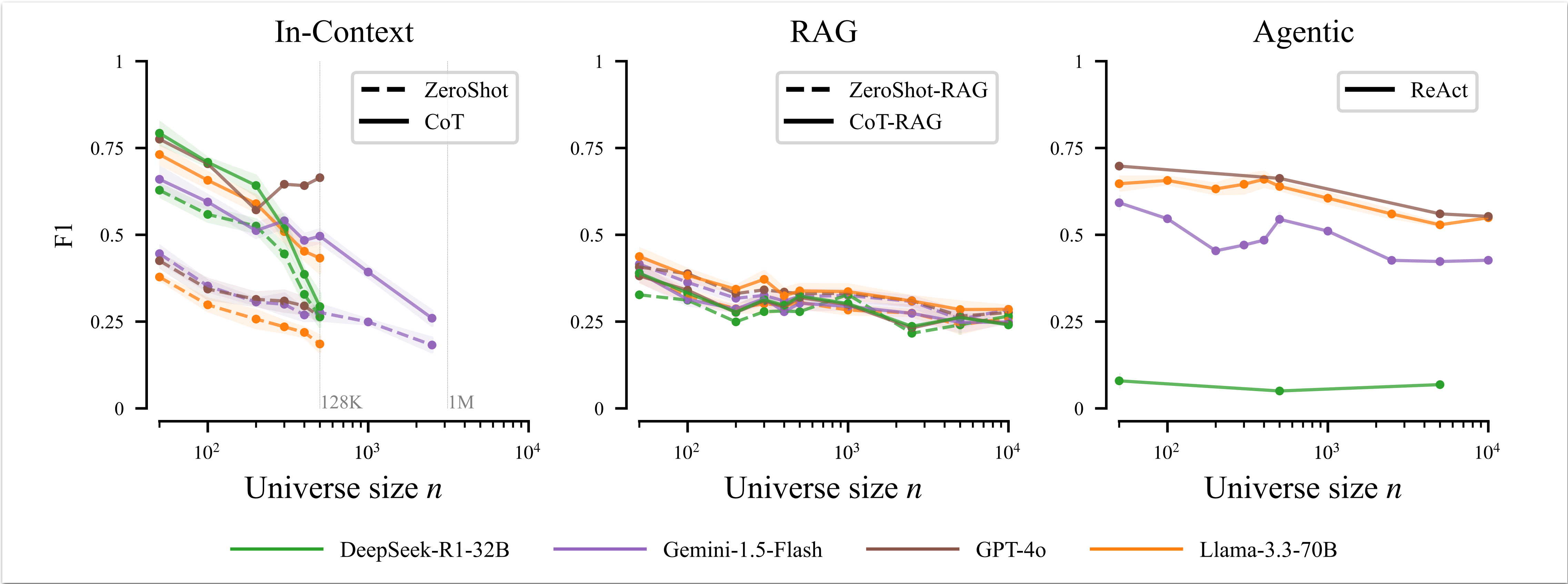
Evaluating Reasoning



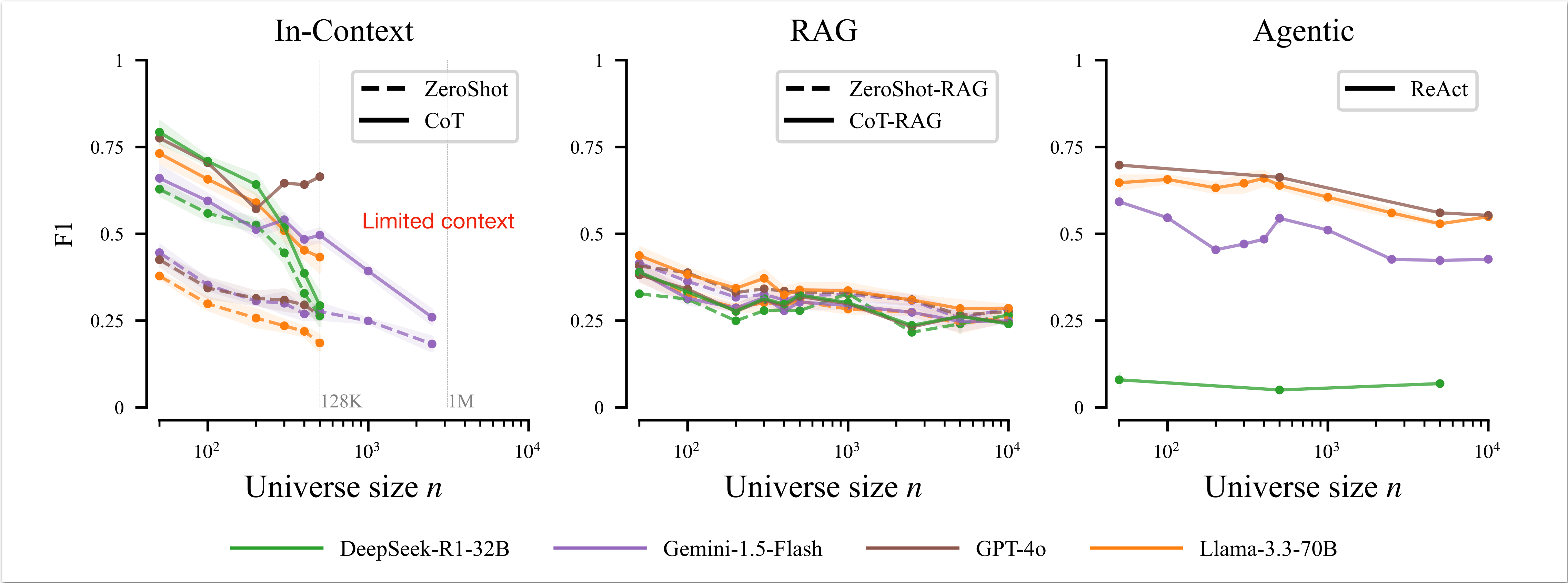
Evaluating Reasoning



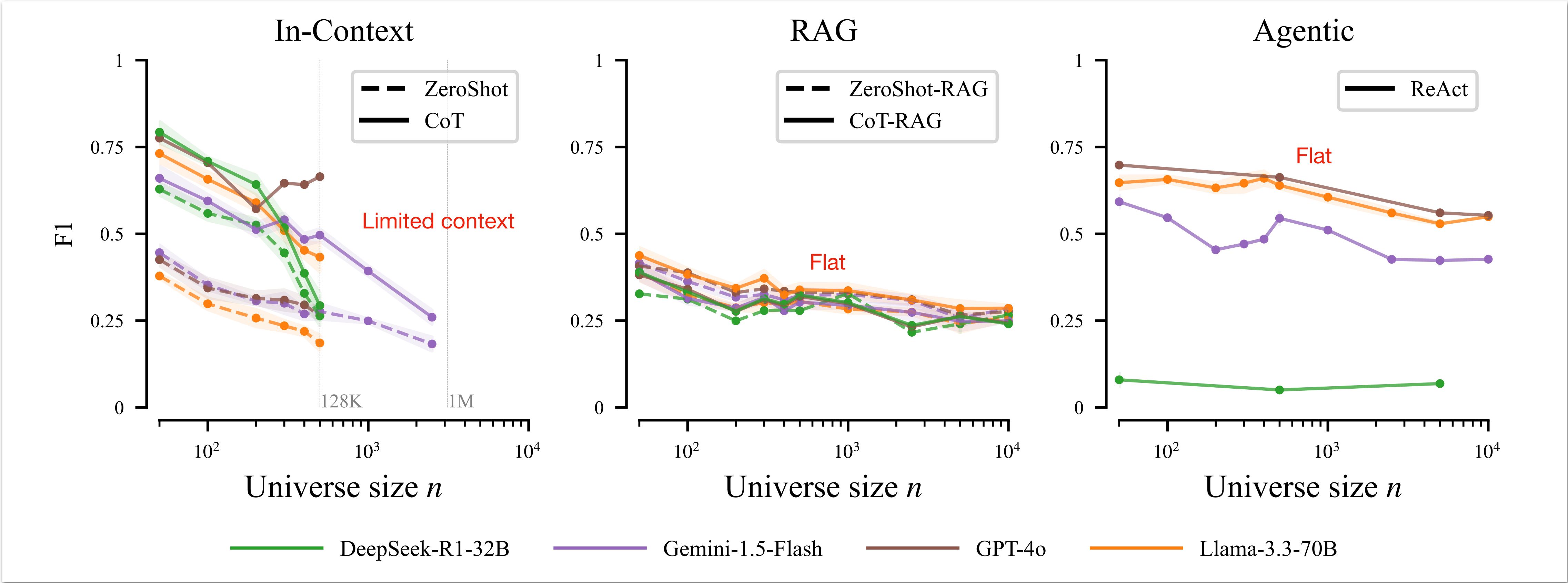
Evaluating Retrieval



Evaluating Retrieval



Evaluating Retrieval



Qualitative Results

Who is the great-grandchild of Gene Smock?

Llama-3.3-70B: To find the great-grandchild of Gene Smock, we first need to identify Gene Smock's children and then their children, and finally, their children.

Gene Smock's child is Williams Smock.
Williams Smock's children are Shelli Beltran and...
Shelli Beltran's children are Aida Wang and...
Aida Wang's children are Alfredo Wang...
Then, we look at the children of Alfredo Wang...

Qualitative Results

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✓ Planning

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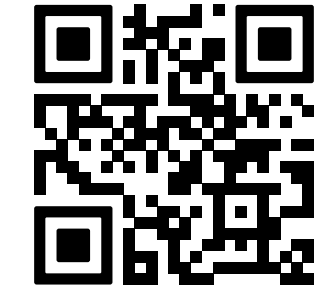
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✗ Execution

Qualitative Results

arxiv.org/abs/2506.20642

✨ Memento ✨



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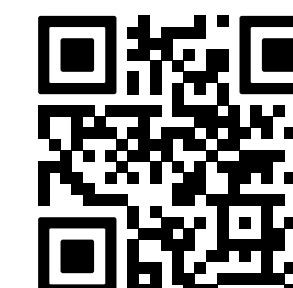
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child("Gene Smock",X),child(X,Y),child(Y,Z)
Target: Z

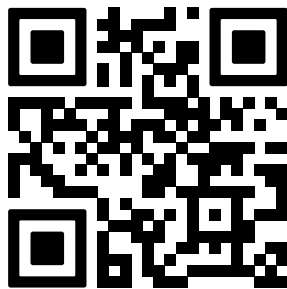
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🌟 Memento 🌟



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 Executor:
X ← Williams Smock

Qualitative Results

arxiv.org/abs/2506.20642

🌟 Memento 🌟



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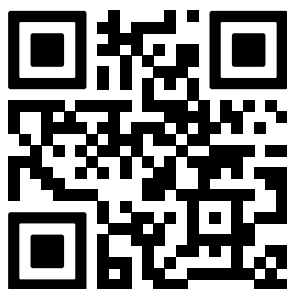
 Executor:
X ← Williams Smock

 Executor:
Y ← Shelli Beltran
Y ← Cynthia Smock
Y ← Stacia Toombs

Qualitative Results

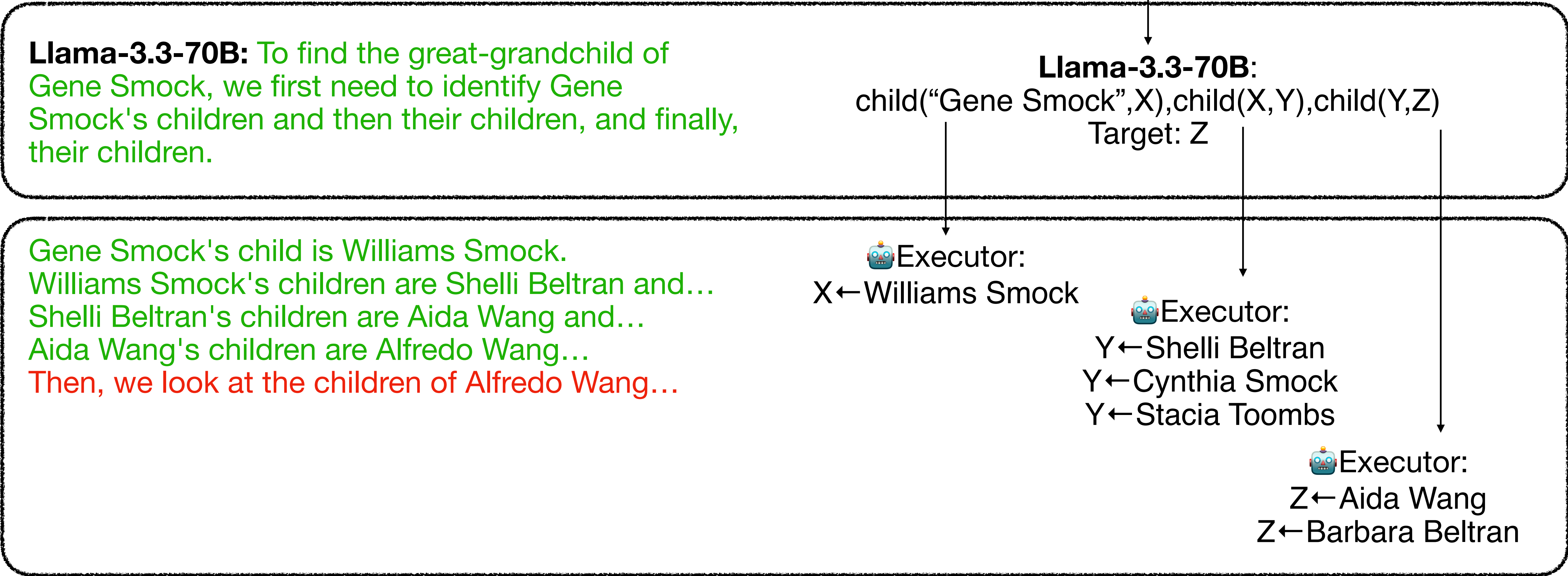
arxiv.org/abs/2506.20642

🌟 Memento 🌟



Who is the great-grandchild of Gene Smock?

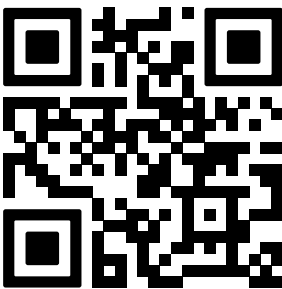
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Qualitative Results

arxiv.org/abs/2506.20642

🌟 Memento 🌟



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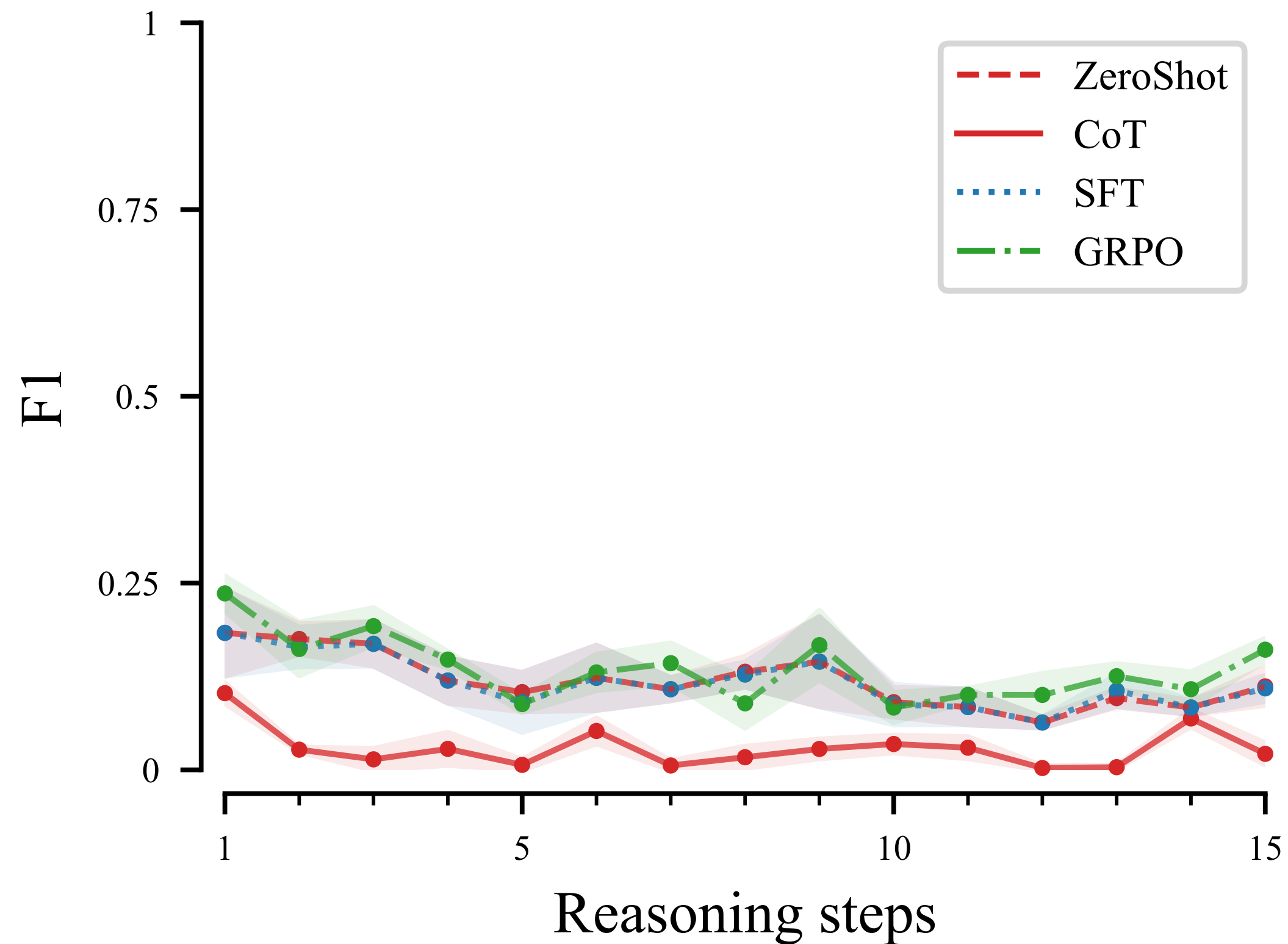
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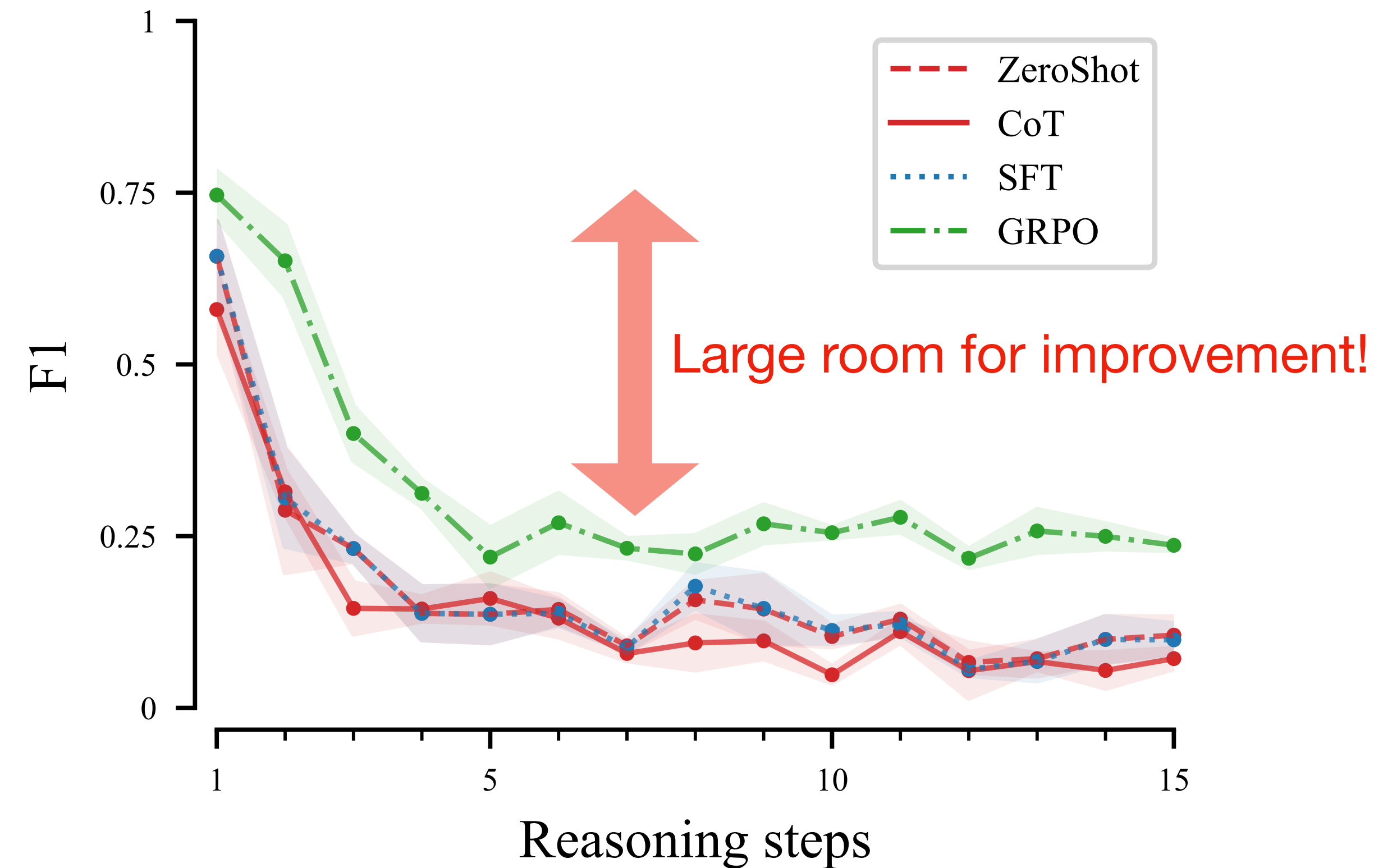
 Executor:
Z ← Aida Wang
Z ← Barbara Beltran

HotpotQA: 49.7 → 53.7
2WikiMultiHopQA: 35.0 → 55.2
MuSiQue: 15.7 → 21.9

Training on PhantomWiki

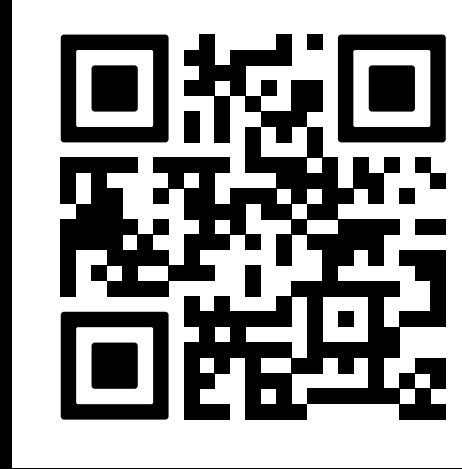


Qwen-2.5-0.5B-Instruct



Qwen-2.5-1.5B-Instruct (LoRA)

Takeaways



arxiv.org/abs/2502.20377

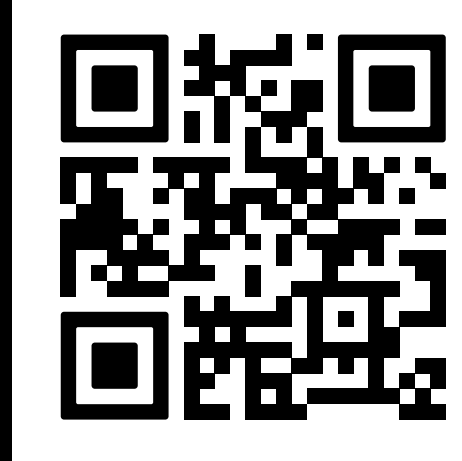


[github.com/kilian-group/
phantom-wiki](https://github.com/kilian-group/phantom-wiki)

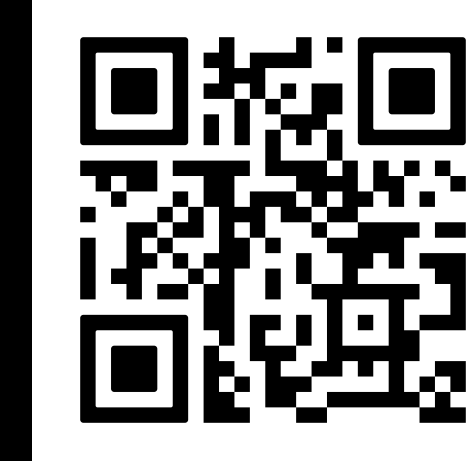
Benefits of PhantomWiki:

- ✓ LLMs have no prior factual knowledge
- ✓ Factually consistent questions and answers
- ✓ Cheap (1M-sized dataset with 4-hop questions takes < 30 CPU hours!)
- 🤖 No LLMs (or humans) are harmed in the process! 🥰

Future Directions



arxiv.org/abs/2502.20377



[github.com/kilian-group/
phantom-wiki](https://github.com/kilian-group/phantom-wiki)

- How can we use Memento to bootstrap PhantomWiki?
- How to scale to 1 billion articles?
- How to utilize intermediate solutions from Prolog for training reasoning models?



LMLM
✨ Pre-training Large
Memory Language Models
with Internal and External
Knowledge ✨

